Agricultural shocks and long-term conflict risk: Evidence from desert locust swarms

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Abstract

Can transitory agricultural shocks affect long-term violent conflict risk? This paper studies this question using data on conflict events and desert locust swarms—localized agricultural disasters—across 0.25° grid cells in Africa and the Arabian peninsula from 1997-2018. A staggered event study approach shows that having been exposed to a locust swarm increases the average annual probability of any violent conflict in a cell by 2.0 percentage points (71%) in subsequent years. Effects are driven by swarms arriving during the main growing season in cells with cropland, with no effects in non-agricultural areas. The largest increases in conflict are delayed and are driven by areas and periods with general increases in insecurity, and particularly by periods with active fighting groups in neighboring cells. I explore income-related mechanisms with a model of occupational choice. Previous studies find persistent effects of locust exposure on measures of household well-being and agricultural profits; I show it also reduces agricultural activity and increases out-migration, but does not affect measures of agricultural productivity. The results are generally not consistent with a persistent decrease in the opportunty cost of fighting in affected areas cause by a permanent income effect of the severe initial shock. Heterogeneity by intensity of the national conflict environment highlights the importance of accounting for context and the feasibility of fighting. Patterns of long-term impacts on violent conflict and heterogeneity by national insecurity are similar for severe droughts, indicating the mechanisms are not specific to locust swarms. These results add further motivation for policies mitigating the risk of agricultural shocks and promoting household resilience and long-term recovery.

JEL codes: D7; O13; Q10; Q54 **Keywords:** conflict; agriculture; desert locusts; natural disasters; Africa

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1 Introduction

A large economic literature explores the impacts on conflict risk of transitory agricultural shocks which do not permanently affect potential land productivity. This is an important policy concern given the prominence of agricultural livelihoods in many of the areas most affected by civil conflict, the threat to agriculture posed by climate change, and the severe economic and human harms of civil conflict (see e.g., Blattman and Miguel, 2010; Fang et al., 2020). Studies of this relationship focus on short-term impacts, and those that analyze shocks to agricultural production are limited in their ability to identify causal mechanisms.[1] This paper analyzes the dynamic long-term impacts of a severe transitory shock to agricultural production—exposure to a desert locust swarm—on conflict risk, and tests for evidence of income-related mechanisms.

Desert locusts are the world's most dangerous and destructive migratory pest (Cressman et al., 2016; Lazar et al., 2016) and effectively constitute an agriculture-specific natural disaster. Climate change is creating conditions more conducive to swarm formation (McCabe, 2021), potentially undoing progress from increased international monitoring and control efforts in recent decades. The arrival of a locust swarm often leads to complete destruction of agricultural production and other vegetation (Symmons and Cressman, 2001; Thomson and Miers, 2002), without the effects on infrastructure or human physiology which may result from weather shocks. Swarm flight patterns create quasi-random variation in the areas exposed to agricultural destruction in a swarm's migratory path, and their migratory nature means that exposure to a swarm does not increase future risk from locusts.These characteristics make locust swarms a useful natural experiment for analyzing how transitory agricultural production shocks affect the risk of conflict.

Using data on the location and timing of desert locust swarm observations from the Food and Agricultural Organization of the United Nations (FAO) and of conflict events from the Armed Conflict Location & Event Data Project (ACLED) and Uppsala Conflict Data Program (UCDP) I estimate a model of conflict at the annual level for 0.25° (around 28×28km) grid cells between 1997-2018 across Africa and the Arabian peninsula.[2] As severe

[1]Several studies find that shocks to agricultural prices increase conflict incidence (e.g., Dube and Vargas, 2013; Fjelde, 2015; McGuirk and Burke, 2020; Ubilava et al., 2022). Impacts on agricultural productivity are speculated to explain the widely-studied relationship between climate or weather shocks and conflict risk (see Burke et al. (2015), Carleton et al. (2016), Dell et al. (2014), Hsiang and Burke (2013), Koubi (2019), and Mach et al. (2019) for reviews), though weather may affect conflict through mechanisms other than agriculture and some studies find results that are not consistent with effects through agricultural productivity (e.g., Bollfrass and Shaver, 2015; Sarsons, 2015).

[2]I include all countries where at least 10 locust swarms are reported during the sample period. Torngren Wartin (2018) estimates short-term impacts of desert locusts on conflict in Africa using similar data. That paper focuses on potential measurement issues around short-term impacts which I discuss in Section 6.4. It

agricultural shocks may have persistent effects on wealth and productivity which could affect conflict risk, I define exposure to a locust swarm as a permanent treatment. I estimate dynamic impacts of swarm exposure using event study designs from the recent literature on difference-in-differences with staggered treatment timing (Borusyak et al., 2024; De Chaisemartin and d'Haultfoeuille, 2024; Roth et al., 2023). I explore patterns in calendar time using an event study of the 2003-2005 major desert locust upsurge, which accounts for 72% of swarm exposure events in the sample period, and estimate average long-term impacts on and heterogeneity by different cell characteristics using two-way fixed effects regressions.

Locust swarms increase the annual probability of any violent conflict event occurring in a 0.25° grid cell by 2.0 percentage points (71%) on average in years after exposure to the swarm, compared to unaffected areas. The event studies show no significant impacts of locust swarms on violent conflict in the year of exposure but increases in all following years up to 12 years after exposure. Impacts are entirely driven by cells with crop or pasture land, and by swarms arriving in crop cells during the main growing or harvest season in particular. I find limited evidence of conflict spillovers outside of exposed cells. The results are robust to a variety of alternative specifications.

I interpret the results and evaluate income-related mechanisms through the lens of a commonly-used model of individual occupation choice between production and conflict (Chassang and Padró i Miquel, 2009; Dal Bó and Dal Bó, 2011). In the model, transitory agricultural shocks affect the short-term risk of conflict by changing both the returns to engaging in agricultural production—the *opportunity cost* mechanism—and the returns to fighting over agricultural output—the *predation* mechanism. I extend the model to allow past agricultural production shocks to affect conflict risk through a wealth or *permanent income* mechanism. Adoption of insurance against negative agricultural shocks is very low in the study area, leading households to undertake costly consumption smoothing strategies and reducing household wealth (financial, physical, and human capital).[3] This wealth effect can decrease long-run productivity, leading to persistent reductions in the opportunity cost of fighting.

Long-term conflict risk does not increase uniformly, with the largest effects on violent conflict risk coming 7-12 years after swarm exposure. I show that this aligns with the gap between the main locust exposure event in 2003-2005 and the onset of various conflicts

does not consider long-term impacts of locust swarms or mechanisms that are the main contributions of this paper.

[3]See for example Alderman et al. (2006), de Janvry et al. (2006), Dercon (2004), Dercon and Hoddinott (2004), Dinkelman (2017), Fafchamps et al. (1998), Hallegatte et al. (2020), Hoddinott (2006), Hoddinott and Kinsey (2001), Maccini and Yang (2009), and Townsend (1995) on coping strategies LMIC agricultural households use to respond to uninsured shocks being largely uninsured and their impacts on household wealth/assets, including long-term consequences.

in the sample countries caused by the Arab Spring, various civil conflicts, and the spread of terrorist organizations. Long-term impacts on violent conflict are concentrated in more insecure periods, and particularly where there are groups actively engaged in violent conflict in the surrounding area. Populations affected by swarms may lack the resources or motivation to form fighting groups to engage in conflict, while fighting groups mobilized following some precipitating event may find areas previously exposed to locust swarms to be easier targets or fertile ground for recruitment. The results thus show that severe agricultural shocks need not cause the onset of new conflicts and highlight the importance of the conflict environment in determining effects of shocks on conflict incidence.

To test whether effects are specific to locust swarms I also test impacts on conflict risk of exposure to severe drought, measured using monthly Standardized Precipitation and Evapotranspiration Index (SPEI) data. I find large significant increases that are also driven by locations and periods with a more insecure conflict environment, indicating the same mechanism underlies the effects of both types of agricultural shocks. The long-term increases in conflict risk imply that analyses defining shock exposure as temporary and estimating short-term impacts using fixed effects (the main method in studies of climate or agricultural shocks and conflict) are misspecified. I show that such analyses result in downward-biased estimates of the short-term impacts of both locust swarms and severe drought on violent conflict, affecting the policy implications.

I find limited evidence in support of an opportunity cost mechanism in either the short or long term. Locust swarms have no significant effect on violent conflict in the year of exposure and a small effect the following year despite this being the period when the opportunity cost effect from reduced agricultural productivity should be highest. This is true for measures of both output conflict and conflict over territory or factors of production, and while the risk of protests does immediately increase significantly, the magnitude is small. Affected populations may lack the resources to fight and be less of a target for predatory attacks in the short-term, and may also prefer to migrate rather than fight.

In the long-term, a permanent income mechanism persistently reducing the opportunity cost of fighting would predict observable decreases in measures of economic activity following swarm exposure. I find no significant long-term effects on the Normalized Difference Vegetation Index (NDVI), a proxy for agricultural production calculated using MODIS satellite imagery, or on Demographic and Health Survey (DHS) AReNA database measures of survey cluster average crop yields, indicating to permanent decrease in agricultural productivity. I do observe significant long-term decreases in crop area cultivated and quantity harvested together with increases in out-migration indicating some transitions away from agricultural work, but the magnitudes are small. Evidence of a permanent income mechanism is therefore mixed, and should be explored further in household-level analyses.

This paper makes several contributions to the literature. First, while a relationship between climate and conflict has been repeatedly demonstrated the mechanisms driving this impact are not fully understood (Burke et al., 2024; Mach et al., 2020). Weather shocks affect a variety of economic and social outcomes in addition to reducing agricultural labor productivity and agricultural output (Dell et al., 2012, 2014; Mellon, 2022), but much of the literature has emphasized an opportunity cost mechanism to explain increases in conflict risk.[4] The results of this paper indicate that the opportunity cost of fighting mechanism alone cannot explain dynamic impacts of locust swarms and drought on conflict risk over time and highlight the importance of the predation mechanism and its influence on opportunities for fighting.

Second, I analyze long-term impacts of an economic shock on conflict to further our understanding of the drivers of conflict (Bazzi and Blattman, 2014; Blattman and Miguel, 2010; Collier and Hoeffler, 1998; Dube and Vargas, 2013; Grossman, 1999; Hodler and Raschky, 2014; McGuirk and Burke, 2020; Miguel et al., 2004), and the role of agricultural production in particular (Crost et al., 2018; Harari and La Ferrara, 2018; Iyigun et al., 2017; McGuirk and Nunn, 2021; Von Uexkull et al., 2016). Studies of the impacts of agricultural shocks on conflict have focused on the short-term.[5] I consider the possibility of long-term impacts through a permanent income mechanism, and test for this channel by examining long-term impacts of desert locust swarms on measures of agricultural productivity and economic activity. I find limited support for this mechanism, but find significant increases in long-term conflict risk following locust and drought exposure which suggest that studies focusing on short-term impacts of severe economic shocks may be misspecified if they define the shock 'treatment' as temporary.

Third, I add to a broader literature on the impacts of environmental shocks and natural disasters. Many papers have explored how environmental shocks can have persistent effects on poverty and well-being (Baseler and Hennig, 2023; Carter and Barrett, 2006; Carter et al., 2007; Lybbert et al., 2004), but these mechanisms have not been related to conflict risk. More generally, the evidence on long-term impacts of disasters such as hurricanes and

[4]Little attention is given to the predation mechanism in the climate-conflict literature. Studies showing evidence of opportunity cost and predation mechanisms in agriculture have primarily explored impacts on conflict risk of changes in global prices of agricultural goods (e.g., Dube and Vargas, 2013; Fjelde, 2015; McGuirk and Burke, 2020) rather than shocks to local agricultural production. McGuirk and Nunn (2021) is an exception, analyzing impacts of drought on conflict between pastoralists and farmers.

[5]Crost et al. (2018) and Harari and La Ferrara (2018) estimate effects of weather shocks on conflict 1 and 4 years afterward. Iyigun et al. (2017) is an exception, considering long-run effects on conflict risk of a *positive and permanent* agricultural productivity shock from the introduction of the potato to the Eastern Hemisphere. To my knowledge, no study has explored long-term impacts on conflict risk of a transitory negative shock to agricultural production.

droughts is limited, inconclusive, and focused on a small number of outcomes (see Botzen et al. (2019) and Klomp and Valckx (2014) for reviews). I study how the impacts of desert locusts swarms—an extreme shock to agricultural production akin to a natural disaster—on conflict risk evolve over time and test whether patterns are consistent with a permanent income mechanism.

Finally, this paper also contributes to a small literature studying economic impacts of desert locusts beyond immediate crop destruction and costs of control operations (see e.g., Thomson and Miers, 2002), and to a slightly larger literature on the long-term impacts of agricultural pests (Baker et al., 2020; Banerjee et al., 2010). The range of many agricultural pests is expanding due to climate change and globalization, and though locust outbreaks have become less frequent in recent decades due to increased monitoring desert locusts are ideally situated to benefit from climate change (McCabe, 2021; Qiu, 2009; Youngblood et al., 2023). Several papers find locust swarm exposure adversely affects long-term productivity and human capital (Conte et al., 2023; De Vreyer et al., 2015; Le and Nguyen, 2022; Linnros, 2017; Marending and Tripodi, 2022), but I am not aware of any study considering the impacts of a pest shock on conflict. The impacts of locust swarms on long-term agricultural productivity and conflict risk should be considered in determining policy around desert locust prevention and control.

The remainder of the paper is organized as follows. Section 2 provides background on desert locusts and summarizes the literature on agricultural shocks and conflict. Section 3 presents a model of how agricultural shocks affect occupational choice and the decision to fight over time through income-related mechanisms. Section 4 describes the data used in the analyses and Section 5 outlines the empirical approach. Section 6 presents the results for the impacts of locust swarm exposure on violent conflict, tests potential mechanisms in light of the model, and compares impacts of exposure to locust swarms and severe drought. Section 7 discusses the results and Section 8 concludes.

2 Background

2.1 Desert locusts

Desert locusts (*Schistocerca gregaria*) are a species of grasshopper always present in small numbers in desert 'recession' areas from Mauritania to India.[6] They usually pose little threat to livelihoods but favorable climate conditions in breeding areas—periods of repeated rainfall

[6] Additional detail on desert locusts is included in Appendix B. Any time I use 'locusts' in this paper I am referring exclusively to desert locusts.

and vegetation growth overlapping with the breeding cycle—can lead to exponential population growth. Unique among grasshopper species, after reaching a particular population density locusts undergo a process of 'gregarization' wherein they mature physically and begin to move as a cohesive unit (Symmons and Cressman, 2001), with adult winged locusts forming large mobile swarms. When swarms migrate away from breeding areas and affect multiple countries this is referred to as an outbreak or upsurge. Climate change is expected to increase the risk of locust swarm formation and upsurges, as desert locusts can easily withstand elevated temperatures and the increased frequency of heavy rainfall events can create conditions conducive to population growth (McCabe, 2021; Qiu, 2009; Youngblood et al., 2023).

Locust swarms vary in density and extent, but the average swarm includes around 50 locusts per m^2 and can cover tens of square kilometers, including billions of locusts (Symmons and Cressman, 2001). About half of swarms exceed 50km^2 in size (FAO and WMO 2016). The size of swarms is what makes them so destructive. A small swarm covering one square kilometer consumes as much food in one day as 35,000 people and the median swarm consumes 8 million kg of vegetation per day (Food and Agriculture Organization of the United Nations (FAO), 2023), without preference for different types of crops (Lecoq, 2003. The arrival of a swarm can lead to the total destruction of local agricultural output (Symmons and Cressman, 2001; Thomson and Miers, 2002). For example, during the 2003-2005 locust upsurge in North and West Africa, 100, 90, and 85% losses on cereals, legumes, and pastures respectively were recorded, affecting more than 8 million people and leading to 13 million hectares being treated with pesticides (Showler, 2019). Over 25 million people in 23 countries were affected during the most recent 2019-2021 upsurge and damages were estimated to reach \$1.3 billion (Green, 2022), with control efforts—including treating over 2 million hectares with pesticides—estimated to have prevented over \$1 billion in damages (Newsom et al., 2021).

Households use a variety of measures to cope with short-term food security and livelihood effects of locust outbreaks. In addition to seeking help from social networks and food aid, households commonly report selling animals and other assets, consuming less food, sending household members away, taking loans and cutting expenses, and consuming seed stocks as coping strategies (Thomson and Miers, 2002). A swarm exposure shock therefore represents a shock to income and household wealth as well as a shock to agricultural productivity in the year of exposure.

An important characteristic of desert locust swarms that is also useful econometrically is that they are migratory. Locusts swarms fly 9-10 hours per day, generally downwind, and easily move 100km or more in even with minimal wind (FAO and WMO 2016). The migratory path of locust swarms tends to bring them to breeding areas in time for seasonal breeding. Conditional on being in the migratory path during an upsurge, swarm flight patterns create quasi-random variation in exposure as some areas in the flight path are flown over and spared any damages.[7]

Locusts live 2-6 months and swarms continue breeding and migrating until dying out from a combination of migration to unfavorable habitats, limited vegetation in breeding areas, and control operations (Symmons and Cressman, 2001). Knowledge of locust breeding patterns and swarm flight characteristics inform efforts to predict locust swarm formation and movements, but forecasts remain highly imprecise (Latchininsky, 2013). Even given such information, farmers have no proven effective recourse when faced with the arrival of a locust swarm (Dobson, 2001; Hardeweg, 2001; Thomson and Miers, 2002). The only current viable method of swarm control is direct spraying with pesticides, which can take days to have effects as well as being slow and costly to organize and requiring robust locust control infrastructure (Cressman and Ferrand, 2021). Farmers in affected areas report viewing locust swarms as an unpredictable natural disaster that is the government's responsibility to address (Thomson and Miers, 2002). Extensive locust monitoring and control operations are conducted in countries at regular risk from locust swarms. These are insufficient to prevent all upsurges but can help limit their spread and damages.

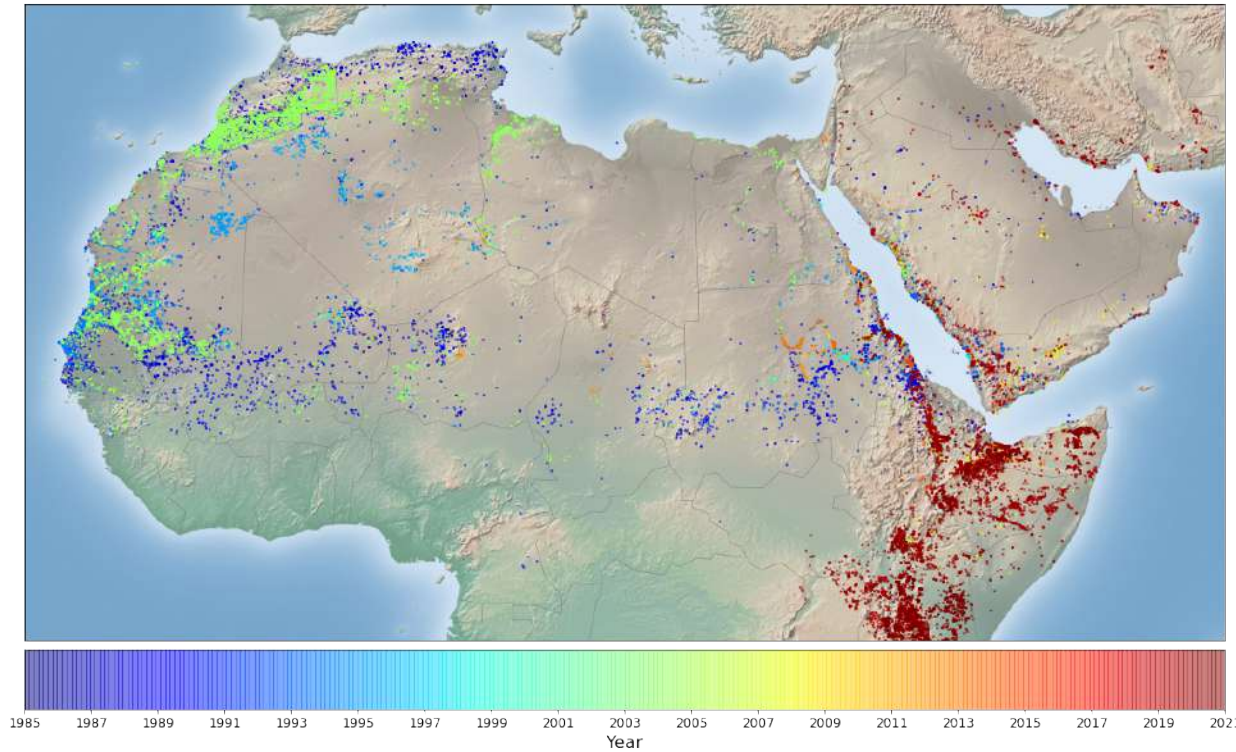
Figure 1 displays the locations of desert locust swarm observations recorded in the FAO Locust Watch database from 1985 (the first year they were recorded) to 2021, for the area of interest for this study.[8] As illustrated by the figure, locust swarms are not observed with any regularity over time and only in a few countries have locations been exposed to more than one swarm. Which locations are exposed to a locust swarm during an upsurge depends on which breeding areas fostered initial swarm formation and on wind patterns in the months following swarm formation. The countries affected by the 2003-2005 upsurge (in green in Figure 1), which originated from multiple small outbreaks in Summer 2003 in the Western Sahel, are not the same as those exposed to the 2019-2021 upsurge (in red), which originated in the southern Arabian Peninsula.

The characteristics of desert locust swarms make them a useful natural experiment for analyzing the long-term impacts of agricultural production shocks on the risk of conflict. First, the timing of upsurges and patterns of swarm flight create quasi-random temporal and local variation in swarm exposure. The arrival of a swarm also does not change future risk (Figure B3 shows this empirically), so the direct shock to agricultural production is

[7] Figure B4 illustrates the local and temporal variation in exposure to swarms for the area around Mali.

[8] Desert locust swarms also affect other countries in the Middle East and South Asia, but not during the time period of this study.

Figure 1: FAO Locust Watch swarm observations by year



Note: Map created by author using locust swarm records in the FAO Locust Watch database.

transitory. Second, the arrival of a swarm is effectively a locally and temporally concentrated natural disaster where all crops and pastureland are at risk (Hardeweg, 2001), but other aspects of the economy are importantly unaffected. Temperature and rainfall shocks may affect infrastructure or physiology as well as agricultural production. Locust swarms may have some psychological effects, but are well-suited for studying income-related mechanisms linking agricultural shocks and conflict. Third, the level of damage to agriculture from swarms and lack of tools for farmers to prevent damages imply severe reductions in agricultural production. Decreased wealth following such a catastrophic shock may be more likely to persist and affect labor productivity in following seasons, increasing the potential for long-term impacts on conflict through the opportunity cost channel.

2.2 Agricultural shocks and conflict

A growing literature explores the impacts of climate or weather on conflict (see Burke et al. (2015, 2024), Carleton et al. (2016), Dell et al. (2014), Hsiang and Burke (2013), Koubi (2019), and Mach et al. (2019) for reviews), primarily analyzing impacts of deviations of rainfall or temperature from local historical norms with some also looking at droughts. Most studies find that weather shocks increase short-term conflict risk, with the recent Burke et al. (2024) meta-analysis finding a mean increase in the risk of intergroup conflict of 2.5% for a one standard deviation adverse change in climate and a median effect of 5%, with more consistent effects of temperature increases than of adverse rainfall realizations.These results have important implications for conflict risk as climate change increases the frequency and severity of weather shocks.

The majority of papers in the climate-conflict literature looking at low- and middle-income countries (LMICs) focus on income-related mechanisms, following early work in Miguel et al. (2004). Arguments typically follow the models in Chassang and Padró i Miquel (2009) and Dal Bó and Dal Bó (2011), discussing how weather affects agricultural labor productivity and therefore the opportunity cost of engaging in conflict for agricultural producers. Studies frequently use variation in effects by land cover or timing relative to the growing season to show support for this mechanism (e.g., Caruso et al., 2016; Crost et al., 2018; Gatti et al., 2021; Harari and La Ferrara, 2018; McGuirk and Nunn, 2021; Von Uexkull, 2014). Although the evidence is generally consistent with an opportunity cost mechanism, reviews of the climate-conflict literature agree that the mechanisms remain unclear and deepening insight into them is highlighted as a priority for future climate-conflict research in Burke et al. (2024) and Mach et al. (2020). Weather affects the economy and society through multiple channels besides agricultural production (Dell et al., 2012, 2014; Mellon, 2022), and studies have pointed to physiological, psychological, and infrastructural effects of weather shocks as also helping to explain impacts on conflict (Baysan et al., 2019; Burke et al., 2024; Carleton et al., 2016; Chemin et al., 2013; Dell et al., 2014; Hsiang and Burke, 2013; Sarsons, 2015; Witsenburg and Adano, 2009). An advantage of analyzing the effects of locust swarm exposure is that physiological and infrastructural effects should be limited—though psychological effects may be important—allowing a cleaner identification of the importance of income-related mechanisms.

Income mechanisms have been discussed and tested in the literature on the drivers of conflict more generally including studies of agricultural shocks not affecting production. Several studies have shown that plausibly exogenous changes in prices of agricultural commodities affect the risk local conflict in areas producing the affected goods (Bazzi and Blattman, 2014; Dube and Vargas, 2013; Fjelde, 2015; McGuirk and Burke, 2020; Ubilava et al., 2022). A recent literature explores how the onset of harvest season in agricultural areas affects economic incentives to fight (Guardado and Pennings, 2021; Hastings and Ubilava, 2023). These studies illustrate different ways in which agricultural shocks affect both the opportunity cost of fighting related to agricultural labor productivity and the returns to predatory capture of agricultural output. In some cases the opportunity cost mechanism appears to dominate (Bazzi and Blattman, 2014; Dube and Vargas, 2013; Fjelde, 2015; Guardado and Pennings, 2021), while in others the predation mechanism is decisive (Koren, 2018; Ubilava et al., 2022), sometimes within the same context (Hastings and Ubilava, 2023; McGuirk and Burke, 2020). Studies identifying results consistent with predation or banditry emphasize the role of armed groups such as militias or insurgents which coordinate such attacks, which indicates that the broader 'conflict environment' may create heterogeneity in the impacts of shocks to agriculture on conflict risk.

The majority of studies on agricultural shocks and conflict focus on impacts within the same time period, with a few exceptions. Crost et al. (2018) and Harari and La Ferrara (2018) find that growing season weather shocks have inconsistent effects on conflict in the same year, but consistently increase conflict risk the following year. Harari and La Ferrara (2018) finds some persistence of impacts up to 4 years afterward. Crost et al. (2018) argue that lagged effects on conflict could be due to storage and savings offsetting effects in the same year. This appears inconsistent with effects driven by reduced opportunity costs as the largest impacts on agricultural labor productivity should be realized in the same season as the rainfall shock. To my knowledge, only Iyigun et al. (2017) consider how a permanent agricultural productivity shock impacts conflict in the long term. They find that introducing potatoes to Europe, the Near East, and North Africa led to a large and persistent reduction in the risk of conflict in subsequent centuries by comparing changes in areas with different suitability for potato cultivation. A paucity of evidence on long-term impacts is a limitation of the literature on natural disasters more generally (Botzen et al., 2019), and particularly in low-income countries (Baseler and Hennig, 2023).

3 Model

The standard models discussed in the literature on agricultural shocks and conflict are models of occupational choice (as in French and Taber, 2011; Heckman and Honore, 1990; Roy, 1951) where actors allocate their labor between productive activities and fighting.[9] In particular, Chassang and Padró i Miquel (2009) develop a bargaining model of conflict where groups allocate labor to crop production or fighting over land, and Dal Bó and Dal Bó (2011) model individuals choosing between a labor-intensive sector, a capital-intensive sector, and an 'appropriation' sector fighting over output. Other mechanisms, such as effects of shocks on physiology, psychology, infrastructure, or institutions may also be important (Burke et al., 2024), but for the purpose of this paper I follow these models in testing income-related mechanisms.

[9] Becker (1968) uses a similar setup to model interpersonal conflict such as theft.

The models illustrate how agricultural shocks affect the risk of conflict through two main, opposing, mechanisms. First, negative shocks such as low prices or drought reduce the returns to agricultural labor. This means the *opportunity cost* of fighting is lower: producers have less to lose by engaging in conflict. At the same time, lower agricultural prices or output reduce the returns to *predation*: bandits or looters have less to gain from fighting. These mechanisms are not unique to agricultural shocks, as they are also discussed in earlier work on the economic drivers of conflict more generally (Collier and Hoeffler, 2004; Grossman, 1999). For transitory agricultural shocks, the value of output available to capture falls but the value of factors of production—land in particular—is less affected. In line with this, the literature generally finds that the opportunity cost mechanism dominates for shocks that temporarily reduce agricultural returns, increasing conflict risk.

Prior research models transitory agricultural shocks as having only temporary effects on conflict. But a transitory income shock can also reduce permanent income when consumption smoothing decreases assets. Most agricultural households in developing countries lack insurance and have constrained access to credit. Strategies to smooth consumption following an income shock, such as selling animals and other assets, taking loans, reducing food, health, and education spending, and sending members away reduce household physical and human capital (e.g., de Janvry et al., 2006; Dercon and Hoddinott, 2004; Dinkelman, 2017). The resulting reductions in wealth mean temporary shocks can have persistent impacts on productivity (Dercon, 2004; Hallegatte et al., 2020; Hoddinott, 2006; Karim and Noy, 2016). In the context of an occupational choice model of conflict, this wealth or *permanent income* effect increases the long-term risk of conflict by reducing the long-term opportunity cost of fighting.

I present a streamlined model of occupational choice as in Chassang and Padró i Miquel (2009) and Dal Bó and Dal Bó (2011) including this long-term permanent income mechanism to build intuition and generate hypotheses about the effect of agricultural shocks on conflict. In the model, individuals in each time period allocate one unit of labor L to either agricultural production, non-agricultural work, or violent conflict to maximize total net income I. Returns to all activities are affected by individual and location characteristics X such as land quality, level of education, and fighting ability.

Net returns to agricultural production $F^A(L^A, S, W, X)$ are affected by agricultural shocks S, with $\frac{\partial F^A}{\partial S} < 0$ and $\frac{\partial^2 F^A}{\partial S^2} < 0$. A larger S therefore reduces agricultural labor productivity—the *opportunity cost mechanism*. Agricultural production also depends on wealth W with $\frac{\partial F^A}{\partial W} > 0$, where wealth broadly includes human, physical, and financial capital. Wealth in period t is weakly increasing in income I from activities in period $t-1$. As agricultural shocks decrease income, this creates a relationship between past agricultural shocks

S_{t-s} and agricultural production in period t, where $s \in [1, \tau]$ for some τ. We can write $F_t^A = F^A(L_t^A, S_t, W_t(\{S_{t-s}\}_{s=1}^{\tau}), X_t))$, with $\frac{\partial F_t^A}{\partial S_{t-s}} < 0$ causing a wealth or *permanent income mechanism*.

Net returns to non-agricultural work $F^N(L^N, X, W)$ are based on the most productive activity available outside of own agricultural production. The highest returns available depends on individual and location characteristics X and wealth W. As a simplifying assumption, I suppress the direct dependence of non-agricultural returns on S. Returns to non-agricultural work thus set a lower bound on how far the opportunity cost of fighting may fall following a negative agricultural shock. F^N will be weakly smaller for individuals primarily engaged in agriculture that experienced a past agricultural shock due to the permanent income mechanism.

Individual i can also decide to engage in predatory violent conflict attacking a set of potential targets J near i that are feasible to attack within the time period. The potential net returns $F^C(L_i^C, X_i, \{I_j, W_j, X_j\}_{j\in J})$ depend on the incomes (production output), wealth (factors of production), and characteristics of the individuals in J. Agricultural shocks S_j for individuals $j \in J$ engaged in agriculture affect i's returns to fighting by decreasing the income available to capture: $\frac{\partial F_t^C}{\partial S_{j,t}} < 0$. This is the *short-term predation mechanism*. Past agricultural shocks to individuals $j \in J$ will reduce both the output and the factors that i can capture through the permanent income mechanism, meaning $\frac{\partial F_t^C}{\partial S_{j,t-s}} < 0$. This is the *long-term predation mechanism*.

The probability of success and costs of fighting depend on characteristics $X_{i,t}$ and $X_{j,t}$—some targets will be farther away or have better fighting ability. An important variable in $X_{i,t}$ is whether there are active fighting groups the individual could join, as this will reduce the costs of fighting for the individual and increase the probability of successfully capturing returns. Costs of fighting are incurred with certainty and include economic, social, and emotional costs as well as risk of injury or death. These costs make fighting sub-optimal for most individuals in most time periods. In practice, individuals are unlikely to engage in violent conflict alone, as such fighting generally involves organized armed groups which recruit members and pay them a wage or share of the returns from victory (Collier and Hoeffler, 2004; Grossman, 1999).

The individual's problem in period t can be presented as choosing their labor allocation $L_{i,t}$ to maximize income $I_{i,t}$ given some current and past shock realizations S_i, S_j. For simplicity and intuition I ignore uncertainty in returns and suppose that decisions are made (or equivalently, updated) after the agricultural shocks in the period are realized.

$$
\begin{aligned}
\max_{L^A_{i,t},L^N_{i,t},L^C_{i,t}} I_{i,t} =& F^A(L^A_{i,t}, S_{i,t}, W_{i,t}(\{S_{i,t-s}\}^{\tau}_{s=1}), X_{i,t}) + F^N(L^N_{i,t}, W_{i,t}(\{S_{i,t-s}\}^{\tau}_{s=1}), X_{i,t}) \\
&+ F^C(L^C_{i,t}, X_{i,t}, \{I_{j,t}, W_{j,t}(\{S_{j,t-s}\}^{\tau}_{s=1}), X_{j,t}\}_{j\in J})
\end{aligned}
$$

$$
\text{subject to } L^O_{i,t} \in \{0,1\}, \quad F^O(0,.) = 0, \quad \text{and } \sum_O L^O_{i,t} = 1 \text{ for } O \in \{A,N,C\}
$$

$$
\frac{\partial F^A}{\partial S_{i,t}} < 0; \quad \frac{\partial F^A}{\partial S_{i,t-s}} < 0; \quad \frac{\partial F^C}{\partial S_{j,t}} < 0; \quad \frac{\partial F^C}{\partial S_{j,t-s}} < 0
$$

This yields

$$
\begin{aligned}
L^C_{i,t} = 1 \text{ iff } & F^C(1, X_{i,t}, \{I_{j,t}, W_{j,t}(\{S_{j,t-s}\}^{\tau}_{s=1}), X_{j,t}\}_{j\in J}) \\
& \geq \max(F^A(1, S_{i,t}, W_{i,t}(\{S_{i,t-s}\}^{\tau}_{s=1}), X_{i,t}), F^N(1, W_{i,t}(\{S_{i,t-s}\}^{\tau}_{s=1}), X_{i,t}))
\end{aligned}
$$

In words: actor i chooses to engage in violent conflict if the net returns from fighting exceed their opportunity cost: the highest net returns they could receive from choosing another occupation.

Conflict occurs in the locations of the individuals being attacked. In this paper I analyze conflict at the level of grid cells which contain many individuals. Although I will define a common shock measure at the cell level, cell size is chosen to ensure there is variation in exposure within cells. I test the sensitivity of the results to using larger grid cells to capture spillovers of conflict outside the areas affected by agricultural shocks.

The effect of an agricultural shock on the decision to fight in the same time period is ambiguous, particularly if there is a strong positive correlation between shocks over space, as in most agricultural shocks. When the shocks are correlated, larger $S_{i,t}$ will decrease i's returns to agricultural production in the same period but also be associated with a decrease in output available to capture from nearby targets for predatory attacks. This makes conflict over output (i.e., banditry) less attractive, but has minimal effect on the returns to conflict over territory or factors of production assuming the shock is transitory.[10] These offsetting effects will be less important for shocks that vary more over space. In the case of this paper, I analyze impacts of swarm exposure at the level of 28×28km grid cells, such that the median locust swarm would only affect around 6% of cell area creating meaningful local variation in shock exposure.

The long-term effects of past agricultural shocks $S_{i,t-s}$ on the decision to engage in conflict

[10] I focus on transitory agricultural shocks which do not have a permanent direct effect on local agricultural productivity. Transitory shocks may have some effect on the returns to fighting over factors if they affect individuals' ability to productively utilize factors or if they affect expectations about future productivity. Shocks that have direct permanent productivity effects, for example through soil erosion or other land degradation, would have larger effects on the returns to capturing factors of production.

in period t also involve offsetting mechanisms. Because of impacts on wealth as a result of consumption smoothing, the returns to both agricultural production and to fighting will be lower than before the shock in affected areas relative to unaffected areas, though higher than in the period of the shock. As with short-term impacts of an agricultural shock, long-term impacts should be smaller for conflict over output than over factors of production assuming their expected returns are not much affected by the transitory agricultural shock.

The permanent income effect is likely to be particularly strong following desert locust swarm exposure due to the severity of the income shock. If the permanent income mechanism is important, this should be observable in long-term impacts on measures of economic activity and agricultural productivity.

Since violent conflict is not the norm in most locations and periods, it implies the returns are generally low. While there is evidence that agricultural shocks motivate the formation of fighting groups and cause the onset of new violent conflict immediately following the shock (Harari and La Ferrara, 2018; McGuirk and Burke, 2020), it is not clear whether lower productivity in the long-term after a shock would lead fighting groups to form in the absence of some precipitating event. This implies that long-term impacts on conflict risk should be greater in insecure periods when armed groups are already mobilized around particular causes, which reduces the costs of fighting and increases the probability of capturing returns. Armed groups may also perceive areas affected by past shocks as vulnerable targets for attacks if the permanent income mechanism reduces their ability to defend themselves.

To summarize, the model informs a set of testable predictions for the impacts of a transitory agricultural shock on conflict risk:

1. If the opportunity cost mechanism dominates, the risk of violent conflict should increase in the year of shock exposure.
2. If the predation mechanism offsets the opportunity cost mechanism, this should attenuate short-term effects on measures of conflict over output but not for conflict over factors.
3. If the permanent income mechanism drives long-term effects on conflict risk through reductions in the opportunity cost of fighting, we should observe fairly stable or at least non-increasing effects in conflict risk following the shock. This assumes that on average households do not become worse off over time following the initial shock.
4. If predation is an important mechanism, the long-term risk of violent conflict should increase by more in periods when fighting groups are active, increasing the expected net returns to fighting.
5. If a transitory but severe shock has long-term effects through a permanent income

mechanism, this should be observable through long-term reductions in measures of productivity.

4 Data

The Locust Watch database (FAO 2022) reports observations of desert locust swarms as well as smaller concentrations of locusts from 1985 to the present. I consider only data on locust swarms, which pose the greatest threat to agriculture and whose flight patterns create local variation in exposure. The Locust Watch data include latitude, longitude, and date of swarm observations. Locust observations are recorded by national locust control and monitoring officers on the ground, but incorporate reports from agricultural extension agents, government officials, and other sources. Local farmer scouts are also often trained in locust monitoring and reporting (Thomson and Miers, 2002).

A concern might be that locust reporting is correlated with violent conflict. Showler and Lecoq (2021)—which I refer to as 'SL2021' in analyses below—review how insecurity has affected national and international desert locust control operations from 1985-2020 across countries where locusts are active. They mention Chad, Mali, Somalia, Sudan, Western Sahara, and Yemen as countries with areas where insecurity has constrained locust control operations in certain periods since 1997. This concern is the focus of Torngren Wartin (2018)'s analysis of the impact of locusts on conflict, which uses similar data but focuses on the short-term, modeling locusts as temporary shocks.

Insecurity is likely less of a constraint for locust monitoring than for control operations. FAO locust monitoring guidelines discuss conducting aerial surveys and using reports from local scouts, agricultural extension agents, security forces, and other sources (Cressman, 2001), which would allow reporting even in insecure areas. The Locust Watch data includes observations of locust swarms even in countries and periods where Showler and Lecoq (2021) indicate control operations have not been possible. For example, the authors mention that control operations in Western Sahara have been largely infeasible due to Polisario activity over the whole sample period, but 166 swarms have been recorded there in 9 different years from 1996-2018. None of the monthly FAO locust swarm bulletins published during the 2003-2005 upsurge—the major locust event in the sample period—mention issues related to insecurity affecting locust monitoring efforts. The share of cells within 50km of a locust swarm observation in a given year that have reports of both violent conflict and a locust swarm in the cell is 27% in the set of countries Showler and Lecoq (2021) indicate pose challenges for locust control, similar but below the 34% in all other countries.

Missing swarm observations in high-conflict areas would bias my estimates downward by including in the control group areas exposed to locusts with likely higher future levels of conflict, as conflict risk is serially correlated. I test the sensitivity of the results to excluding the countries listed in SL2021 as potential locations of locust swarm under-reporting, to randomly imputing 'missing' locust swarms near the locations of reported swarms, and to randomly imputing 'missing' locust swarms only in such areas also experiencing violent conflict.

Data on conflict events come primarily from the Armed Conflict Location & Event Data Project (ACLED) database (Raleigh et al., 2010). The database records the location, date, actors, and nature of conflict events globally starting from 1997 by compiling and validating reports from traditional media at different levels, from institutions and organizations, from local partners in each country, and from verified new media sources. The analysis focuses on events categorized by ACLED as "violent conflict," which includes battles, explosions, and violence against civilians. I also test impacts on protest and riot events recorded by ACLED and on larger-scale violent conflicts from the Uppsala Conflict Data Program (UCDP; Sundberg and Melander, 2013) and distinguish between conflict that does and does not involve state actors, as these types of conflict will involve different mechanisms. The UCDP database goes back to 1989 and only records conflicts involving at least one "organized actor" and resulting in at least 25 battle-related deaths in a calendar year. The ACLED database has no organized actor or minimum death threshold requirements. McGuirk and Burke (2020) characterize UCDP events as more likely to represent conflict over territory and factors of production, and I follow them in constructing a measure of output conflict (i.e., banditry) using ACLED records of violence against civilians, rioting, and looting.

I collapse the data to a raster grid with annual observations for cells with a $0.25°$ resolution (15 arcminutes, approximately $28{\times}28$km). Analyzing impacts at this spatial level reduces potential measurement error about the specific areas affected by swarm and conflict events and allows me to leverage local variation in swarm presence created by their flight patterns. Nearly all swarms will be contained within $0.25°$ cells ($\sim$784km^2), except those near cell boundaries. I test for robustness to analyzing data at the level of $0.5°$ and $1°$ cells, which will also capture potential spillovers from swarm exposure (McGuirk and Nunn, 2021). In each cell and year I measure whether any locust swarm and conflict event was recorded.

I determine the country and highest sub-national administrative level in which each cell centroid lies using country boundaries from the Global Administrative Areas (2021) database v3.6. I use sub-national boundaries to create a set of 285 regions, all of which include at least 32 individual grid cells. These regions are either existing sub-national administrative units or combinations of adjacent units within the same country. I cluster standard errors at the level of these regions.

Given the role of weather in desert locust biology, its importance in determining agricultural production, and the well-documented relationship between weather shocks and conflict, all analyses control for local weather to isolate the impact of locust swarm exposure. I measure total annual precipitation (in mm) and maximum temperature (in °C) using high-resolution monthly data from WorldClim available through 2018.[11] I use CEDA monthly Standardized Precipitation and Evapotranspiration Index (SPEI) data for all of Africa from 1981-2016 at a 5km resolution (Peng et al., 2019) to create a measure of severe drought exposure, which I define as at least 4 consecutive months in a year where the SPEI is below -1.5 (with values from -1 to 1 indicating normal conditions). I define the maximum Normalized Difference Vegetation Index (NDVI) at the annual level for each cell using 16-day 1km satellite imagery from MODIS (Didan, 2015).

I also incorporate raster population data for every 5 years from CIESIN, 2018, linearly interpolating within cells between years where the population is estimated, and raster data on land cover in 2000 from CIESIN, giving the share of land cover in each cell that is cropland and pasture (Ramankutty et al., 2010). I combine the land cover data with cropland mapping of Africa from 2013-2014 (Xiong et al. (2017)) to identify cells with any cropland during the study period. I include additional cell characteristics from the PRIO-GRID dataset (Tollefsen et al., 2012), assigning all 0.25° cells the values for the 0.5° cell in which they are located.

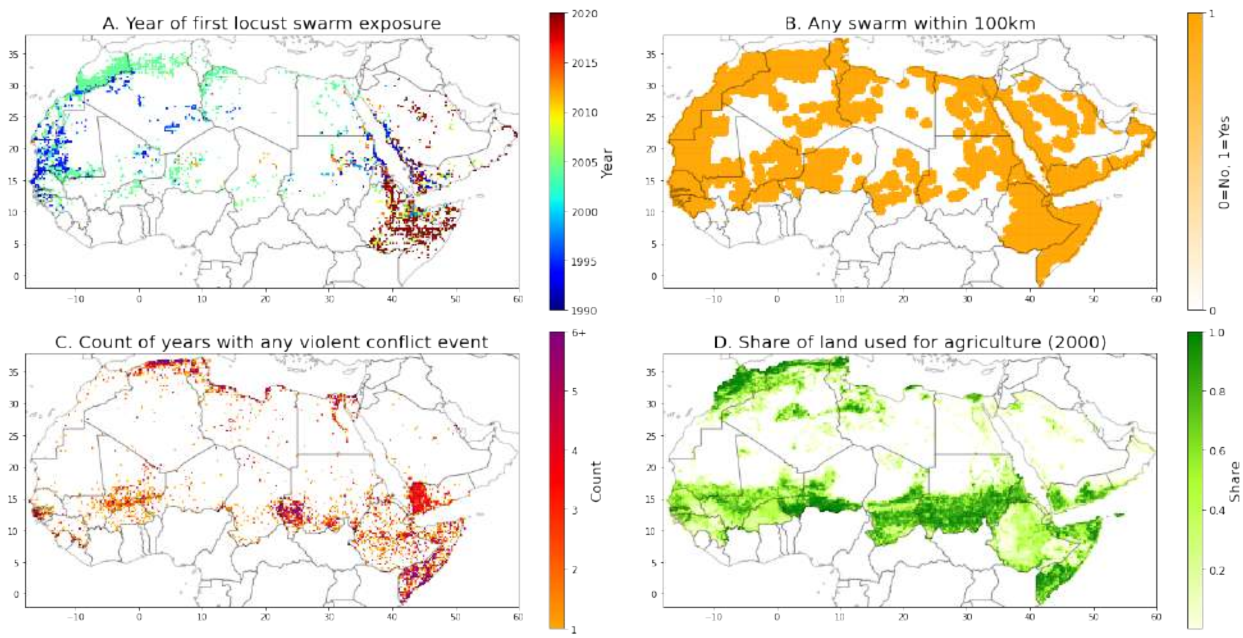
For the analysis of mechanisms, I incorporate data on net migration and agricultural production. Net migration at the 5 arcminute resolution for 2000-2019 come from Niva et al. (2023). Measures of agricultural production as well as nightlight intensity come from the DHS AReNA database (IFPRI 2020), which includes geolocated data at the level of household survey clusters for 40 surveys from 9 countries in the study sample conducted between 1992 and 2018.

Since ACLED records conflicts beginning in 1997 and the main weather data are available until 2018, the analysis sample includes observations from 1997 to 2018. I restrict the analysis to countries with at least 10 locust swarm observations in this period. These countries include all of North Africa, most of the Arabian Peninsula and West Africa, and the Horn of Africa. The resulting analysis sample covers 22 years across 25,553 cells, for a total of 561,726 observations. Figure 2 visualizes swarm exposure, violent conflict incidence, and agricultural land cover for the sample countries. Summary stats are included in Table A1.

A locust swarm is recorded in the Locust Watch database between 1985-2021 for 12% of cells (Table A1 Panel B), and 62% of cells are within 100km of a locust swarm report

[11] CRU-TS 4.03 (Harris et al., 2014) downscaled with WorldClim 2.1 (Fick and Hijmans, 2017). I test sensitivity to measuring rainfall using CHIRPS and temperature using ERA5 (Hersbach et al., 2019) to account for satellite-based weather measurement error (Josephson et al., 2024).

Figure 2: Swarm exposure, violent conflict incidence, and land cover in sample countries



Note: Land used for agriculture includes crop land and pasture land. Panel D shows most clearly which countries in West, Central, and East Africa are excluded from the study sample.

(Figure 2 Panel B). For each cell, I identify the first year after 1989 in which a locust swarm is recorded (Figure 2 Panel A), and define a cell as exposed to a locust swarm in each following year and not exposed in all other years or if no locust swarm is ever observed.[12] Locations where locust swarms are observed in more than one year (9.8% of exposed cells) are not distinguished from those where they are observed only once. Cells first exposed to a swarm from 1990-1997 (in dark blue in Figure 2 Panel A) are considered treated during the entire sample period and therefore do not inform the analyses, while cells first exposed to a swarm after 2018 (in red) are considered not treated during the sample period. Just over seven percent of cells are first exposed to a swarm during the sample period, including 5.3% exposed during the 2003-2005 upsurge (in teal).

Thirteen percent of cells experienced at least one violent conflict event, in 3.4 different years on average (Table A1 Panel B). The large majority of conflict events are recorded after 2010, as illustrated by Figure 2 Panel C. The risk of any violent conflict is fairly low from 1997-2010 before increasing significantly over the remainder of the sample period (Figure 4 Panel A). Part of this increase may related to changes in ACLED data collection methods, though I observe a similar increase in the UCDP data. More directly, the increase

[12]A major locust upsurge occurred from 1985-1989. Treatment timing for cells where another locust swarm was recorded is based on the first year after this period that a locust swarm was recorded, while cells where no other locusts swarms were recorded are considered not treated during the 1997-2018 study period. The results are robust to basing treatment timing on the first year any swarm is recorded in a cell in the Locust Watch database.

corresponds with the timing of the Arab Spring movements, the spread of Islamic militant groups, and multiple civil wars and separatist movements in the sample countries.

Over half the cells (57%) in the sample had agricultural land in 2000: 56% have pasture land while 31% have crop land. Across all cells, the mean share of land allocated to agriculture is 23% (Figure 2 Panel D, Table A1 Panel A), with 18% pasture land and 5% crop land. Given that locust swarms should affect outcomes through agricultural destruction, I test for heterogeneity in impacts by land cover.

5 Empirical approach

I estimate the causal impacts of locust swarm exposure on conflict using a difference-in-differences approach allowing for long-term effects of this transitory agricultural shock. I estimate both static average impacts using two-way fixed effects (TWFE) models and dynamic impacts over time using event study approaches. The TWFE linear probability models take the form:

$$Conflict_{ict} = \alpha + \beta Exposed_{ict} + \delta X_{ict} + \gamma_{ct} + \mu_i + \epsilon_{ict} \tag{1}$$

where i indexes cells, c countries, and t years. $Conflict$ is a dummy variable for observing any conflict event and *Exposed* is a dummy variable for having been exposed to a locust swarm. The primary specifications focus on impacts on violent conflict using the ACLED data. I consider effects on other outcomes in tests of the impact mechanisms. Analyzing conflict as a binary variable at an annual level reduces potential measurement error and is the main approach in the literature. γ_{ct} are country-by-year fixed effects, and μ_i are cell fixed effects. X_{ict} is a vector of time-varying controls at the cell level including rainfall, maximum temperature, and population in the main specifications. Standard errors (SEs) are clustered at the sub-national region level (285 clusters) to allow for correlation in the errors within nearby areas over time.[13]

I estimate dynamic impacts over the 12 years before and after locust swarm exposure using staggered treatment event study models, including the same fixed effects and clustering as in Equation 1.[14] These event study approaches deal with concerns with TWFE estimators

[13]This is likely more restrictive than necessary and will lead to a conservative interpretation of the results. Patterns of statistical significance are largely unchanged when using two-way clustered errors at the year and region level and using Conley (1999) Heteroskedasticity and Autocorrelation-Consistent (HAC) SEs allowing for spatial correlation over 100 and 500km and serial correlation over 0 or 10 time periods, following Hsiang (2010)'s approach.

[14]The main estimates do not include controls but results are robust to including the same controls as in the TWFE analyses.

when there is heterogeneity in treatment effects by time since treatment or across treatment cohorts, which can lead to 'forbidden' comparisons between late- and early-treated groups and negative weighting of effects for certain treatment groups or periods (Goodman-Bacon, 2021). The methods effectively estimate an average treatment effect on the treated in each time period separately for groups exposed in different years, and calculate event study estimates by taking weighted averages of these treatment effects. I primarily present results using the approach developed in Borusyak et al. (2024) but test the sensitivity of results to alternative esimators including Callaway and Sant'Anna (2021), Cengiz et al. (2019), and De Chaisemartin and d'Haultfoeuille (2024). The estimators mainly differ in how they estimate pre-trends; Borusyak et al. (2024) impute counterfactual untreated outcomes for all units and make comparisons against the average over all pre-treatment periods, leader to smoother pre-treatment dynamics (Roth et al., 2023).

To test for heterogeneity in the impacts of swarms, I estimate Equation 1 fully interacting the right-hand side variables with another variable of interest. I test robustness of the results to different controls and fixed effects, restrictions of the analysis sample, cell sizes, and clustering of SEs. Results of robustness tests are included in Appendix C.

The key identifying assumption of this design is that trends in conflict risk would be parallel over time in exposed and unexposed areas within the same country in the absence of locust swarm exposure, after controlling for effects of weather and population. While this assumption is not possible to test directly, I can test its plausibility in two main ways: testing for baseline balance in cell characteristics and testing for parallel trends prior to exposure.

Cells exposed to a locust swarm during the sample period have different baseline characteristics than unexposed cells which are largely consistent with desert locusts rarely being observed in the interior of the Sahara desert (as shown in Figure 2). Exposed cells have larger populations, are closer to capital cities, have a greater share of pasture land and smaller share of barren land, and have lower maximum temperatures (Table A2). These differences remain significant but are smaller when restricting the sample to cells within 100km of any locust swarm from 1996-2021 (a joint test of differences in cell characteristics yields $F = 3.35$ and $p < 0.01$). The differences are largely eliminated when weighting observations by the inverse of the estimated probability of locust swarm exposure, and I test the sensitivity of the results to including these inverse propensity weights (Stuart et al., 2014). In the main analyses I restrict the sample to cells within 100km of any locust swarm and within the range of common support of the estimated probability of swarm exposure across exposed and unexposed cells, and test the sensitivity to alternative restrictions.

Baseline differences are not a concern if they do not affect conflict risk or only affect levels of conflict, but it is plausible that some of the differences by swarm exposure status would affect conflict trends. However, the controls in the empirical specifications should absorb most of these differences. Cell fixed effects control for time invariant cell characteristics that might affect the risk of conflict such as distance to major cities or country boundaries, topography, and agricultural suitability.Country-by-year fixed effects flexibly control for factors varying over time at the country level that might affect conflict risk, such as food price shocks, weather patterns, the policy environment and national economic and social conditions. Importantly, they also control for trends in violent conflict incidence, which increases over the sample period. I also directly control for time-varying characteristics that differ between exposed and unexposed cells: population, precipitation, and temperature.

Further, I find similar probability of violent conflict by swarm exposure in the pre-exposure periods (Figure 3). The main event study estimate shows significantly *lower* risk of violent conflict in exposed cells in one pre-exposure period (9 years before exposure), but the other pre-exposure coefficients are fairly close to 0 and not statistically significant. A joint test of significance of all 12 pre-exposure coefficients yields $p = 0.342$, indicating no differential conflict risk pre-trends.[15] Though this does not preclude the possibility that trends would differ in the years after swarm exposure for reasons unrelated to agricultural destruction, it is an encouraging sign that the parallel trends assumption may be likely to hold.

Another identification assumption relevant to event study designs with staggered treatment timing is the no anticipation assumption: knowledge of future treatment timing does not affect current outcomes (Roth et al., 2023). Populations may expect a higher probability of swarm exposure in years of major upsurges but cannot perfectly anticipate timing of exposure. For example, the FAO Desert Locust Watch publishes monthly forecasts of areas predicted to be at risk of locust swarm exposure but the predictions include a great deal of uncertainty due to unpredictable variation in swarm flight patterns. Consequently, areas forecast to be at risk are generally quite large, the majority of which end up not being affected by locusts.[16]Anticipation may also have limited effects as there are no effective methods of defending vegetation against locust swarms, and farmers in at-risk areas typically describe locust prevention and control as out of their hands and the responsibility of governments (Thomson and Miers, 2002).

[15] Figure C1 shows results using alternative estimators. I find a similar pre-trends pattern using the Cengiz et al. (2019) event study estimator, with point estimates slightly more negative on average. No pre-exposure coefficients are significant using the De Chaisemartin and d'Haultfoeuille (2024) and Callaway and Sant'Anna (2021) methods, but point estimates for periods 8 to 12 years before exposure are more positive, potentially due to not being able to use country-by-year fixed effects in the Stata packages for these estimates.

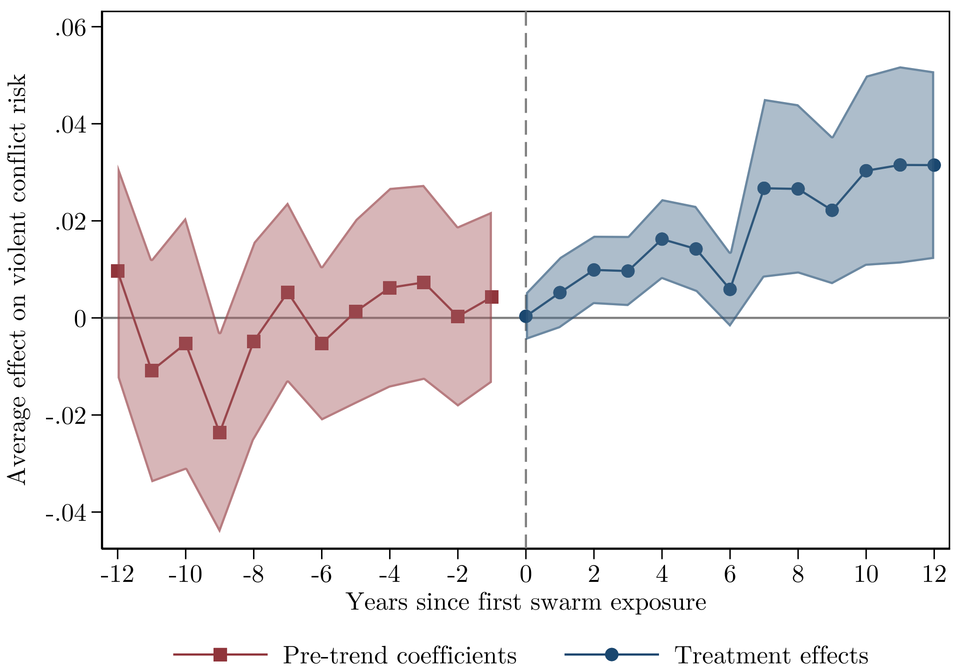
[16] Monthly forecasts during the major upsurge in 2004 on average covered 40.6% of 0.25° cells in sample countries. Despite this, nearly one-quarter of recorded swarms across were not in areas forecast to be at risk in the month the swarm was observed.

6 Results

6.1 Dynamic impacts of swarm exposure on violent conflict

Figure 3 Panel A presents event study estimates of the impacts of swarm exposure on the risk of violent conflict over time using the staggered treatment timing difference-in-differences approach developed in Borusyak et al. (2024). I can reject that the pre-exposure effect is different from zero at a 95% confidence level for one of 12 periods—on average violent conflict is 2.4 percentage points less likely in areas exposed to locust swarms compared to unexposed areas 9 years before exposure. No other pre-exposure coefficient is larger than 0.011 in magnitude, and 7 are positive while 5 are negative. The average pre-exposure difference is -0.001, and I fail to reject that pre-exposure differences are jointly equal to 0 ($p = 0.342$). If anything, the one significant pre-exposure difference being negative suggests exposed cells may have lower baseline risk of violent conflict.

Figure 3: Impacts of exposure to locust swarms on violent conflict risk over time



Note: The dependent variable is a dummy for any violent conflict event. Estimated impacts in each time period are weighted averages across effects for swarm exposure in particular years, calculated using the Borusyak et al. (2024) approach. Time period 0 is the year of first swarm exposure. A joint test that the pre-exposure coefficients equal 0 gives $p = 0.342$. Shaded areas represent 95% confidence intervals using SEs clustered at the sub-national region level. All regressions include country-by-year and cell fixed effects. Observations are grid cells approximately 28×28km by year.

The point estimate for the effect on violent conflict risk in the year locusts arrive is a fairly precise 0. All other estimated treatment effects are positive, and all but the effects in periods 1 and 6 after the year of swarm exposure are statistically significant at a 99% confidence level. The average effect across all post-exposure periods is a 1.9 percentage point increase in the annual risk of any violent conflict event, but effects are not stable over time. The average effect in years 1-6 post-exposure is a 1.0 percentage point increase in conflict risk compared to 2.8 over years 7-12, with highly significant effects over this period even as the standard errors increase with more time since exposure.

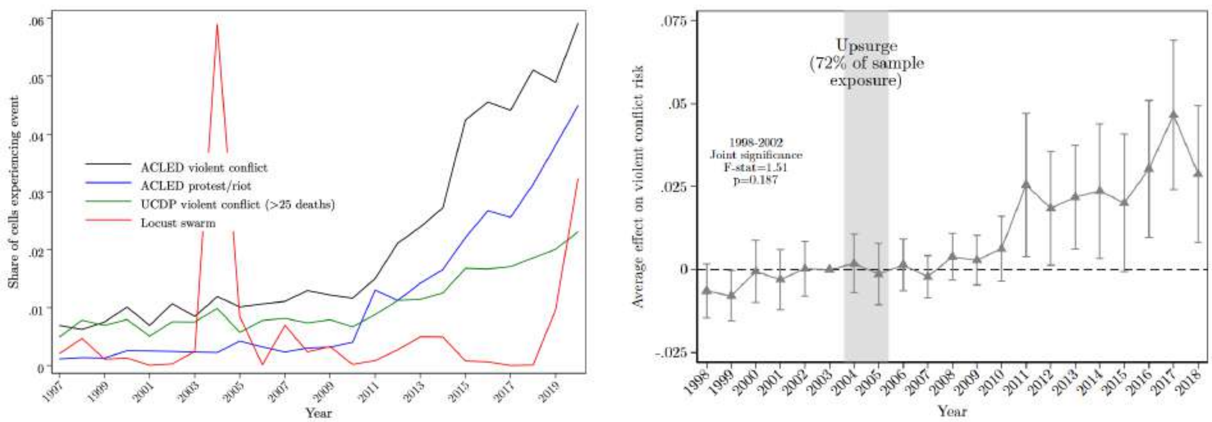
The pattern of results is similar using alternative event study estimators including Callaway and Sant'Anna (2021), Cengiz et al. (2019), and De Chaisemartin and d'Haultfoeuille (2024) (Figure C1), varying the fixed effects and controls, and reducing or increasing the number of pre-exposure effects estimated (Figure C2). Estimated treatment effects are smaller in magnitude and only statistically significant for years 7-12 post-exposure when I include inverse propensity weights based on the estimated probability of being exposed to a swarm during the study period, to account for baseline differences in cell characteristics (Figure C2 Panel D). I observe similar patterns in impacts of exposure over time when collapsing the data to 0.5 or 1° cells, with smaller and marginally significant effects in the first 6 years post exposure and larger and highly significant effects in years 7-12 (Figure C3). There are some positive pre-trends at the 1° level, and estimated effects in larger cells are larger in absolute magnitude but similar in relative magnitude accounting for higher risk of any conflict being observed when considering larger areas. The results are similar when including the full sample of cells and when dropping various geographic regions (Figure C4). Effects of swarm exposure are slightly smaller in magnitude and less precise when dropping countries where Showler and Lecoq (2021) indicate insecurity has limited locust control operations.

Why are there no immediate impacts on the risk of violent conflict and why are the largest impacts delayed? The opportunity cost mechanism alone, acting through immediate effects of swarm exposure on agricultural productivity or through persistent effects on permanent income, would suggest that impacts of swarm exposure on conflict risk should be largest in the short-term and either fall over time as affected areas recover or be fairly stable if households reach a new productivity equilibrium. The fact that long-term impacts of swarm exposure on conflict are not consistent over time and are largest 7-12 years after exposure indicates another mechanism creating heterogeneity in impacts.

An important observation is that the gap between swarm exposure and the largest impacts on violent conflict risks corresponds to the gap between the timing of the main swarm exposure event in the sample period—the 2003-2005 upsurge—and the years when the general risk of conflict increased across the sample countries due to the Arab Spring, the spread of Islamic terrorist groups, and multiple civil wars, as shown in Figure 4 Panel A. Panel B shows that exposure to this upsurge did not significantly increase the risk of violent conflict until 2011, the year of several uprisings related to the Arab Spring, but that effects remain large and statistically significant in subsequent years. I find similar patterns when looking at effects of the upsurge in different countries with different events precipitating the spread of violent conflict (Figure A1).

Figure 4: Changes in conflict environment and impacts of exposure to 2003-2005 locust upsurge on violent conflict risk

A) Swarm and conflict trends B) Event study 2003-2005 upsurge



Note: Panel A shows the share of sample cells experiencing any locust swarm or conflict event by year. Panel B shows results for an event study of exposure to the 2003-2005 desert locust upsurge, with 2003 as the reference period, following Equation 1. The dependent variable is a dummy for any violent conflict event, and treatment is defined as any locust swarm observation from 2003-2005. The bars represent 95% confidence intervals using SEs clustered at the sub-national region level. A joint test that the pre-exposure coefficients equal 0 gives $p = 0.187$. Observations are grid cells approximately 28×28km by year.

As the 2003-2005 locust upsurge accounts for 72% of swarm exposure in the sample period, its effects drive the main event study including all swarm exposure events. Figures 3 and 4 show clearly that exposure does not generally cause the immediate onset of new violent conflicts. Instead, exposed areas appear to be more vulnerable or susceptible to changes in the general conflict environment, implying mechanisms related to the returns to engaging in conflict and not just the opportunity cost of fighting. Significant increases 2-6 years after exposure in the main event study compared to null effects over this period for the 2003-2005 upsurge indicate that later swarm exposure events, which coincided with greater general insecurity, did not have the same delay before increasing violent conflict risk. I return to this below in subsection 6.3 in the analysis of mechanisms.

6.2 Average impacts of swarm exposure on violent conflict

I now estimate average long-term impacts of swarm exposure on the annual risk of violent conflict. Figure 5 shows that on average cells exposed to locust swarms are 2.0 percentage points (pp) more likely to experience any violent conflict in a given year in the period after swarm exposure than cells not exposed. This estimate is very close to the average of the treatment effects in Figure 3—a 1.9pp increase—indicating limited bias in the TWFE estimates from staggered timing of swarm exposure. A 2.0pp increase on the probability of any violent conflict in a year represents an 71% increase over the mean for cells not exposed to locusts.[17]

The average long-term impact of swarm exposure is large compared to the same-year effect of a standard deviation (SD) increase in rainfall relative to cell averages during the sample period, but similar to the effect of a SD increase in maximum temperature. A 1 SD increase in annual precipitation increases the probability of violent conflict in the same year by 0.4pp (14%) compared to 1.9pp (68%) for a 1 SD increase in the maximum annual temperature. The effect of temperature is in the upper end of the distribution of estimates of the impacts of climate on intergroup conflict in Burke et al. (2024)'s meta-analysis, potentially because of the use of maximum temperature and the time period studied. Cell population is also positively associated with conflict risk, with an increase of 10,000 people associated with a 0.9pp (32%) increase in the probability of any violent conflict event in a year.

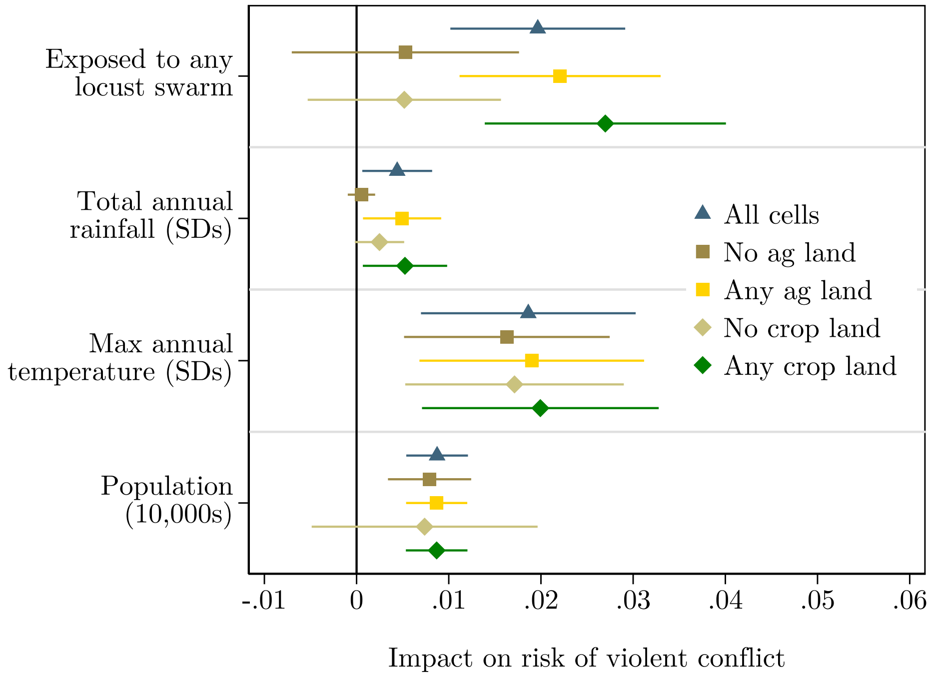
I test for differences in the impacts of swarm exposure and weather by whether a cell has any agricultural (crop or pasture) land or any crop land. Locust swarm exposure in non-agricultural cells (43% of the sample) has no significant effect while in agricultural cells annual violent conflict risk increases by 2.2pp. Locust swarms increase annual violent conflict risk by 2.7pp in crop cells (31% of the sample) compared to no significant effect in non-crop cells (35% of which have pasture land), indicating that impacts on crop production drive most of the effect on conflict risk. Effects of rainfall are driven by cells with agricultural land but not specifically by crop land, while effects of temperature do not vary by land cover, echoing previous work questioning whether agricultural mechanisms explain the relationship between climate and conflict (Bollfrass and Shaver, 2015; Sarsons, 2015). The association between population and conflict risk does not vary significantly with land cover.

To further test that locust swarms represent shocks that affect the economy and society solely through their impacts on agricultural production, I consider heterogeneity in impacts by both land cover and by timing of swarm exposure relative to local crop calendars (**??**). I categorize swarms as arriving during particular stages of the crop production cycle by matching the month in which a swarm is observed to crop calendar information from the PRIO dataset, filling in missing data with country-level crop calendars from The United States Department of Agriculture (USDA) (2022).[18] The off season—between harvesting

[17] The average impact of locust exposure remains statistically significant at the 99% confidence level under other forms of standard error clustering, including two-way clustering at the region and year level and using Conley (1999) SEs allowing for spatial correlation within 100 and 500km and serial correlation over 0 or 10 years (Figure C5). Clustering at the sub-national region level consistently leads to SEs at least as large as Conley SEs allowing for spatial correlation within 500km and serial correlation over 10 years, implying the main SEs I report are conservative and may understate statistical significance of certain estimates.

[18] In countries with different agricultural cycles by crop, I identify the crop activity associated with the

Figure 5: Average impacts of exposure to locust swarms on violent conflict risk by land cover



Note: The dependent variable is a dummy for any violent conflict event observed. Coefficients and 95% confidence intervals are from three separate regressions in each panel: one with no land cover interactions and the other two interacting all right-hand side variables with cell land cover dummies. The coefficients and standard errors for cells with any agricultural (pasture or crop) land and with any crop land are calculated using Stata's *xlincom* command based on the sums of the coefficients for the baseline effect and the interaction term. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level. Results from the regressions are reported in Table A3.

and planting—lasts between 3 and 6 months in most of the sample countries, with an average of slightly over 4 months. I distinguish between swarms arriving during the off-season and planting season (first two months of the crop calendar) when they are unlikely to significantly damage crop production, from swarms arriving the growing and harvesting season when potential damages should be greatest.Figure 6 Panel A presents the timing of locust swarms by region across the sample period. Swarms in cells with crop land are fairly evenly distributed across stages of the crop calendar.

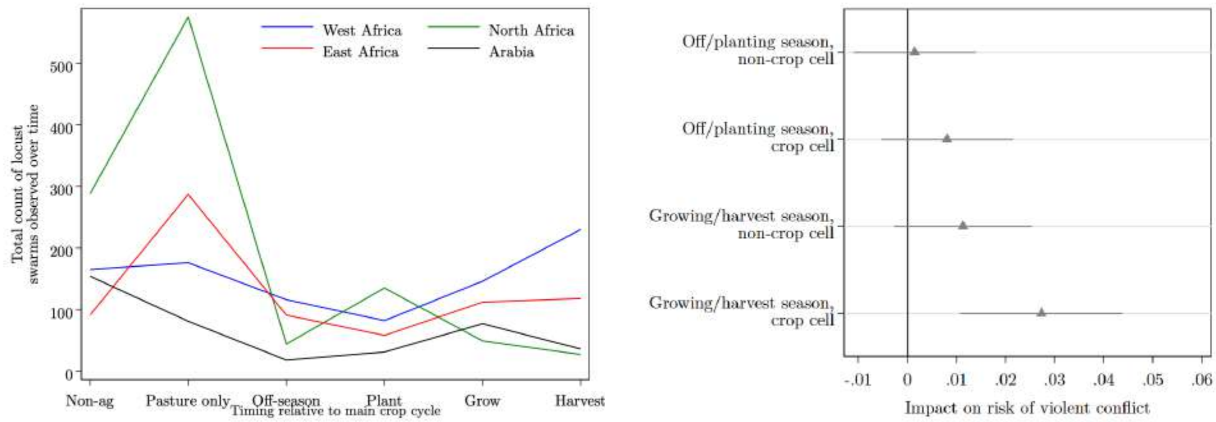
As with the main analysis, I identify the first year a cell was exposed to a locust swarm during a particular season and consider the cell as 'treated' in subsequent years. I then estimate Equation 1 with the two seasonal swarm exposure treatments and fully interacting all variables with a dummy for having any crop land in a cell. Figure 6 Panel B shows that effects are driven by exposure to locusts swarms during the growing or harvest season in crop cells. Exposure to off-season or planting season swarms in crop cells has no significant effect, and the impacts are significantly different ($p = 0.087$). What we might consider 'placebo' swarms in non-crop cells during the off or planting seasons have an estimated effect close to

most commonly grown crops each month.

Figure 6: Impacts of swarm exposure by land cover and swarm timing

A) Variation in swarm timing and location

B) Impacts by swarm exposure timing and location



Note: Panel A identifies the timing of swarm observations based on information on local crop calendars for major crops. Panel B presents coefficients and 95% confidence intervals from a single regression interacting seasonal swarm exposure treatment variables with a dummy for crop land cover with the same fixed effects and controls as in Equation 1. The outcome variable is a dummy for any violent conflict event observed. Observations are grid cells approximately 28×28km by year. Results from the regressions are reported in Table A4.

0, in line with expectations. Growing or harvest season swarms in non-crop cells are close to being significant ($p = 0.113$), suggesting effects through destruction of pasture or other vegetation. The finding relates to studies showing that the impact of weather shocks on conflict risk varies depending on whether the timing of the shock is such that it is likely to decrease agricultural productivity (Caruso et al., 2016; Crost et al., 2018; Harari and La Ferrara, 2018).

The result is also reassuring as it indicates that the estimated impacts are not driven by potential bias in where locust swarms are reported. I further test the potential for missing swarm observations to affect the estimates by simulating how the results change as an increasing share of cell-years where locust swarms are imputed within 100km of a locust swarm report. A particular concern is that swarms may be particularly unreported in insecure areas (Showler and Lecoq, 2021; Torngren Wartin, 2018). This type of measurement error should bias my estimates downward as conflict is serially correlated, and I confirm that estimated effects of swarm exposure increase if I impute 'missing' swarms in areas experiencing violent conflict near existing swarm observations (Figure C6 Panel A). Estimated effects of swarm exposure on violent conflict risk fall if I randomly impute hypothetical missing swarms near the locations of reported swarms. But even if I randomly assign 50% of cells near a swarm report to be exposed to locusts the estimated effect remains economically and statistically significant, with a mean 0.5pp increase in violent conflict risk that is significant at the 95% level in 83% of simulations (Figure C6 Panels B and C). These results strongly imply that the estimated effects of locust swarm exposure are not driven by selection in where swarms are reported.

I test the sensitivity of the results to various alternative specifications and estimate similar impacts of locust swarm exposure on violent conflict risk (Appendix C). Estimated impacts are largely unchanged when varying the set of control variables included, though are larger with no controls and slightly smaller when including year rather than country-by-year fixed effects (Figure C7 Panel A). The standard error is smaller but the coefficient is almost identical when using sub-national region by year fixed effects to identify effects off of more local variation in swarm exposure. The magnitudes of the effect of swarm exposure on violent conflict risk are slightly smaller but remain statistically significant when weighting observations by the inverse of the propensity to have been exposed to a swarm (Figure C7 Panel B), indicating that restricting the sample to areas within 100km of a swarm observation is largely sufficient for identifying a set of control cells to serve as a counterfactual.

Results are similar when including all cells in the sample and when dropping cells in particular geographic regions. This addresses concerns that the long-term impacts on violent conflict may be spurious and due to swarm exposure during the sample period being correlated with factors driving later conflict emergence. For example, dropping North Africa ensures that results are not driven by the Arab Spring and dropping Arabia ensures results are not driven by the civil war in Yemen. The estimated magnitudes are slightly smaller when dropping countries where Showler and Lecoq (2021) report insecurity prevented some locust control operations during the sample period, and when dropping cells that experienced violent conflict during the 2003-2005 locust upsurge which might have prevented swarm reporting (Figure C8 Panel A). Dropping individual years when locust swarm exposure events occurred does not affect the estimates, though the estimate is noisier when dropping the main 2004 exposure event (Figure C8 Panel B).

Absolute effects are larger when collapsing the data to the level of 0.5 or 1° cells, but effects relative to the probability of any violent conflict in unexposed cells are similar (Figure C7 Panel B). Using larger grid cells addresses several potential measurement issues. First, it minimizes the possibility that the area exposed to a locust swarm recorded in a cell exceeds the boundaries of the cell. Second, it reduces concerns about nearby areas that might have been affected by unreported swarms since the entire cell is considered exposed if any swarm is reported within it. Third, it limits the potential for conflict spillovers outside the cell. Downsides to analyzing impacts in a more coarse grid are dilution of treatment intensity (as the share of the cell affected by swarms weakly decreases with cell size) and the loss of local variation in swarm exposure which is important to the identification approach. Larger estimated magnitudes when using larger grid cells despite dilution of treatment intensity indicate potential spillovers of violent conflict outside exposed areas, and test for this directly. I define a spillover exposure treatment as being within 100km of a swarm outside of the cell. I find a fairly precise null average effect of this spillover treatment when controlling for direct swarm exposure, though spillovers from the 2003-2005 upsurge are marginally significant and indicate a 0.7pp average long-term increase in conflict risk for cells within 100km of a swarm during that upsurge (Table C1). Focusing on impacts within 0.25° cells may therefore understate the full effect of swarm exposure on violent conflict risk around affected areas, but the results indicate that most effects are contained within cells which is consistent with swarms only affecting a small portion of cell area.

6.3 Mechanisms

Null effects of locust swarms on violent conflict in the year of exposure are not in line with model predictions based on a dominant opportunity cost mechanism (prediction 1). One possibility is that the predation mechanism is offsetting the opportunity cost mechanism in the short term. If this is the case, the model predicts smaller short-term effects for conflict over output—reduced by the agricultural production shock and therefore decreasing returns to predation—than over factors of production, whose returns are not directly affected by the transitory shock (prediction 2).

I follow McGuirk and Burke (2020) in using reports of violence against civilians, riots, and looting from ACLED as representing conflict over output and violent conflict events reported in the UCDP database as more likely to represent conflict over factors of production. Figure 7 presents event studies for the effects of swarm exposure on risk of these measures of output and factor conflict, and illustrate the difference in impacts. I find null effects on swarms on conflict risk in the year of exposure for both conflict types. If anything the point estimate is small but positive for output conflict but a tight zero for factor conflict, and both are fairly close to zero in the subsequent period. More generally, effects on UCDP conflict are generally not statistically significant including in the period 7-12 years after exposure when the general conflict environment was more insecure. In contrast, effects on conflict targeting civilians are much greater in this period follow that same patterns as effects on any violent conflict in Figure 3.[19] Larger long-term effects on a measure of output conflict than one of factor conflict implies much of increase in conflict stems from banditry or looting rather than civil conflict over control of territory. But the lack of different short term effects are not consistent

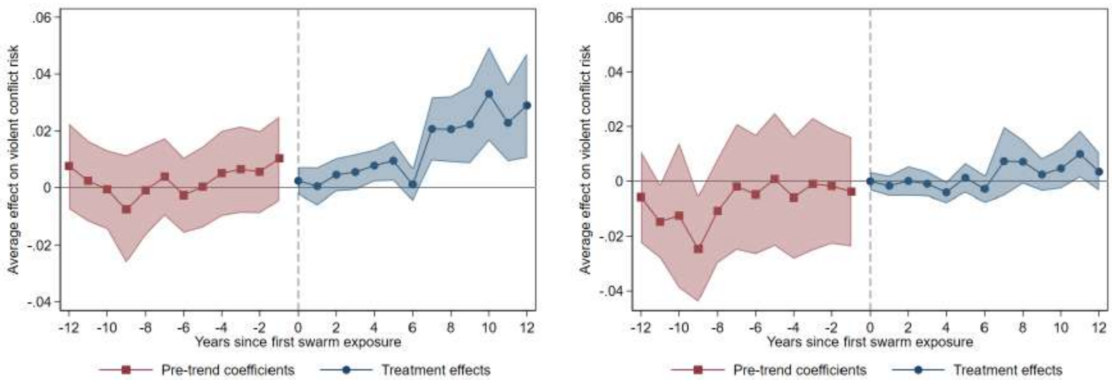
[19] I find significant average long-term increases in conflict risk across multiple types of conflict following exposure to a locust swarm (Table A5). Using ACLED data, estimated effects are similar for violent conflict that does and does not involve any state actor, but are relatively larger for conflict targeting civilians and for protests and riots. Impacts of swarm exposure are smaller in both absolute and relative terms using UCDP reports of violent conflict, which require at least one organized actor and >25 deaths in year. I find no significant effects on fatalities using either ACLED or UCDP data.

with prediction 2 of decreased returns to predation offsetting decreasing opportunity costs of fighting in the short-term, meaning another explanation is needed for the null short-term effects of swarm exposure.

Figure 7: Impacts of swarm exposure on conflict risk over time, by conflict type

A) Any ACLED conflict targeting civilians (output)

B) Any UCDP conflict - organized, >25 deaths (factor)

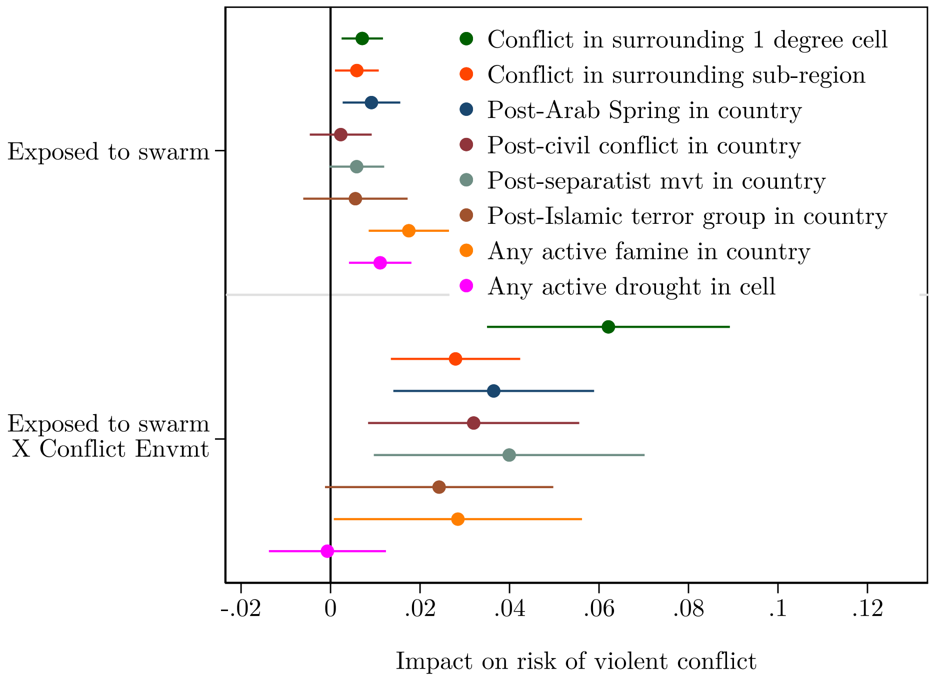


Note: The dependent variables are dummies for any conflict event being observed in a cell in a year, with the conflict type specified in the panel title. Each panel replicates Figure 3 Panel A for a different conflict outcome. See the figure note for Figure 3 for more detail. Shading represents 95% confidence intervals using SEs clustered at the sub-national region level.

Prediction 4 of the model provides a potential alternative explanation: null short term effects may reflect relatively peaceful conditions at the time of the main locust exposure events, limiting the feasibility of fighting and the potential net returns. Figure 4 shows that this was the case for the 2003-2005 locust upsurge, and that conflict did not increase in exposed areas until several years later in the context of a more insecure conflict environment. Heterogeneity in impacts based on the conflict environment could also explain why the results reject prediction 3 of the model. If changes in opportunity costs of fighting under a permanent income mechanism drive long-term impacts of swarm exposure on conflict risk and households do not on average become worse off over time, we would expect effects to be fairly stable over time. Instead, the largest increases in conflict risk are delayed.

To test for heterogeneity in effects of swarm exposure by intensity of the conflict environment, I define a variety of variables indicating the presence of active conflicts at either the local or national level. Figure 8 shows that average impacts are smaller and in some cases not statistically significant in locations and areas not characterized by some form of conflict or insecurity, while effects are significantly larger where there is active conflict, supporting prediction 4 of the model based on a predation mechanism. The largest increase is for areas with concurrent violent conflict in the surrounding 1 degree cell. In these situations, past swarm exposure increases the probability of any violent conflict within the cell by 6.2pp, meaning areas exposed to locusts are more susceptible to the spread of violent conflicts. A variety of factors influence the broader conflict environment and prompted the onset of violent conflict in the sample countries. Impacts of swarm exposure on violent conflict are larger in countries that have experienced Arab Spring uprisings, civil conflict, separatists movements, and Islamic terror attacks.

Figure 8: Heterogeneity in impacts of exposure to locust swarms on violent conflict risk by intensity of conflict environment



Note: The dependent variable is a dummy for any violent conflict event observed. Coefficients and 95% confidence intervals are from separate regressions where I interact Equation 1 with a dummy variable for an indicator of a more intense conflict environment. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level. Results from the regressions are reported in Table A6.

Several mechanisms could explain this heterogeneity. First, these conflicts may further decrease local labor productivity, decreasing the opportunity cost of fighting. If effects of the initial swarm exposure are not sufficient to motivate affected individuals to choose to fight, an additional negative shock could provide the necessary push. In this case I would expect impacts of swarm exposure to be concentrated in periods with other negative weather shocks. But while effects are indeed larger in years when countries are experiencing a famine as shown in the figure, there is no significant difference in effects by whether a cell is experiencing a drought even as these droughts do themselves increase conflict risk (Table A6). Second, areas exposed to locust swarms may be less able to defend themselves from attacks and therefore be targeted by armed groups when these become active. The same mechanisms that would make exposed areas more vulnerable—persistent reductions in wealth—would also make them less attractive targets, however, so it is unclear how the expected returns to predation in these areas would change.

A third possibility relates to the predation mechanism and variation in the expected returns to engaging in violent conflict (prediction 4). Recruitment of fighters to armed groups following the onset of conflict will be easier in areas where opportunity costs of fighting are lower (Collier and Hoeffler, 2004). Individuals in those areas may otherwise not find switching to fighting optimal as violent conflict is generally a collective activity. Without a group with which to fight, the social, emotional, and monetary costs of fighting are likely to be high and the probability of capturing returns low, as discussed in Section 3. A reduction in opportunity costs of fighting following an agricultural shock may thus primarily induce affected individuals to switch to fighting in periods when engaging in fighting is more accessible. Hastings and Ubilava (2023) find evidence of this mechanism in Southeast Asia. They show that the onset of rice harvest in Southeast Asia—a temporary increase in the returns to fighting over agricultural output—only increases violence against civilians in areas with fighting groups active during the growing season. In the absence of such groups, the costs of engaging in predatory conflict during the rice harvest may remain too high relative to the potential returns from fighting without a group.

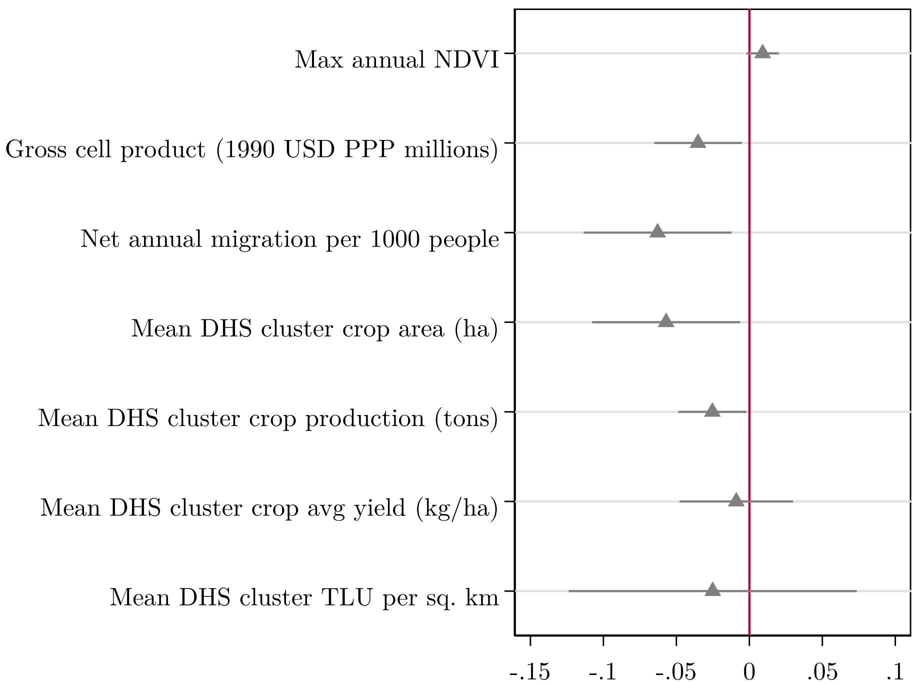
These mechanisms all presuppose persistent effects of swarm exposure on the opportunity cost of fighting through a permanent income mechanism. If this is the case we should observe persistent effects swarm exposure on measures of productivity in affected areas (prediction 5). Other studies have found effects of swarm exposure that could reduce productivity. Most directly, Marending and Tripodi (2022) find that agricultural profits of households in parts of Ethiopia exposed to locust swarms in 2014 were 20-48% lower two harvest seasons after swarm arrival. This indicates that impacts on agricultural productivity are not limited to the year of swarm exposure. Indirectly, several studies show that young children exposed to locust swarms achieve lower educational attainment (De Vreyer et al. (2015)) and have lower height-for-age (Conte et al., 2023; Le and Nguyen, 2022; Linnros, 2017) when they are older. Such human capital effects of swarm exposure could decrease permanent labor productivity.

I present results of tests of average long-term effects of swarm exposure on various measures related to economic activity in Figure 9. I first test for effects on the Normalized Difference Vegetation Index (NDVI), a commonly-used satellite-based measure of vegetation greenness which in crop land can be considered a rough proxy for agricultural production. While I find small but significant decreases in NDVI in the years in which locust swarms are reported (Figure A2), these effects do not persist.[20] On average, locust swarm expo-

[20] An event study analysis of impacts on maximum NDVI in crop cells (Figure A2) shows generally higher NDVI in exposed cells in the years prior to swarm exposure, indicating baseline differences in productivity.

sure has no significant effect on NDVI in subsequent years and the point estimate is small but positive. This result is consistent with locust swarms—a migratory pest—not affecting agricultural productivity fundamentals, but indicates that to the extent they affect later agricultural productivity through effects on household productive assets this is not reflected in measures of NDVI at the cell level.

Figure 9: Average impacts of exposure to locust swarms on measures of economic activity



Note: The figure shows coefficients and 95% confidence intervals from separate regressions of swarm exposure on different outcomes, following Equation 1. All outcomes are normalized so units are standard deviations. NDVI is calculated from MODIS satellite imagery. Gross cell product is from Nordhaus (2006) as included in the PRIO-GRID database. Annual net cell migration is from Niva et al. (2023). DHS cluster data are from the DHS AReNA database (IFPRI 2020). 'TLU' indicates Tropical Livestock Units. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level. Results from the regressions for non-normalized outcomes are reported in Table A7.

I find a significant decrease in gross cell product—estimates of total income at the cell level based on population and output in agricultural and non-agricultural activities from Nordhaus (2006), included in the PRIO-GRID database—following swarm exposure. These estimates are only available for 2000 and 2005 in the sample period, so the results represent the immediate impact of exposure to the 2003-2005 locust upsurge. Total income falls by 8 million USD (in 1990 PPP terms)—nearly 14%—in cells exposed to locust swarms. While this outcome measure is only an estimate of cell-level income, the result indicates large immediate economic impacts in line with research on the agricultural damages from locust swarms. A reduction in income of this magnitude, particularly if concentrated in the

These differences are generally reversed in the post-exposure period, as 9 of 12 point estimates for treatment effects are negative though only 3 are statistically significant. Unexpectedly, I also find significant *increases* in maximum NDVI in two post-exposure periods.

subset of the cell actually affected by locusts, could conceivably reduce household wealth and permanent income.

Niva et al. (2023) estimate annual net migration at the level of 10×10km cells based on sub-national birth rate, death rate, and population data. Using these data, I find that locust swarm exposure decreases net annual migration by 13 people per 1,000 population. This economically and statistically significant effect indicates past swarm exposure drives persistent out-migration from affected areas, which could be consistent with lower labor productivity.

I next test for impacts on measures of household agricultural production from the DHS AReNA database (IFPRI 2020). The samples for these analyses are smaller as they only include cells and years where the DHS is conducted, but they provide a useful test of average household-level impacts. Locust swarm exposure significantly decreases total crop area planted and production in survey clusters in the years following exposure, by 119 ha (5%) and 869 tons (6%), respectively (Table A7). These decreases balance out and result in a null effect on crop yields, and I also find a null and very noisy effect on density of livestock ownership (measured in Tropical Livestock Units).

Taken together the results are not conclusive of an important persistent decrease in labor productivity or wealth in areas exposed to a locust swarm which could reduce the opportunity cost of fighting, as predicted under the model through a permanent income mechanism. NDVI and crop yields do not decrease significantly, indicating agricultural productivity is not reduced in years following a locust swarm on average. Livestock ownership is also not significantly lower, which does not align with a permanent income mechanism based on an initial income shock depleting household assets. It is possible that non-significant average effects on crop yield and livestock ownership mask more concentrated negative effects. The median locust swarm covers around 50 km^2, or 6% of the area of a 0.25 degree cell. Most DHS clusters in cells exposed to locust swarms will therefore not have been affected by any agricultural destruction, so the intent to treat effects I estimate will be biased toward 0.

Significant effects of swarm exposure on crop production and out-migration suggest some households leave agriculture in favor of migrating, in line with reductions in agricultural productivity prompting a shift in occupation. If migration is an attractive alternative to agricultural production for households in areas exposed to locust swarms it may be the case that engaging in violent conflict would also be attractive in some circumstances, particularly in periods with active conflicts around which to mobilize. But additional data, particularly from household surveys, are needed to further test predictions of the permanent income and opportunity cost mechanisms.

6.4 Comparing locust swarms and severe drought

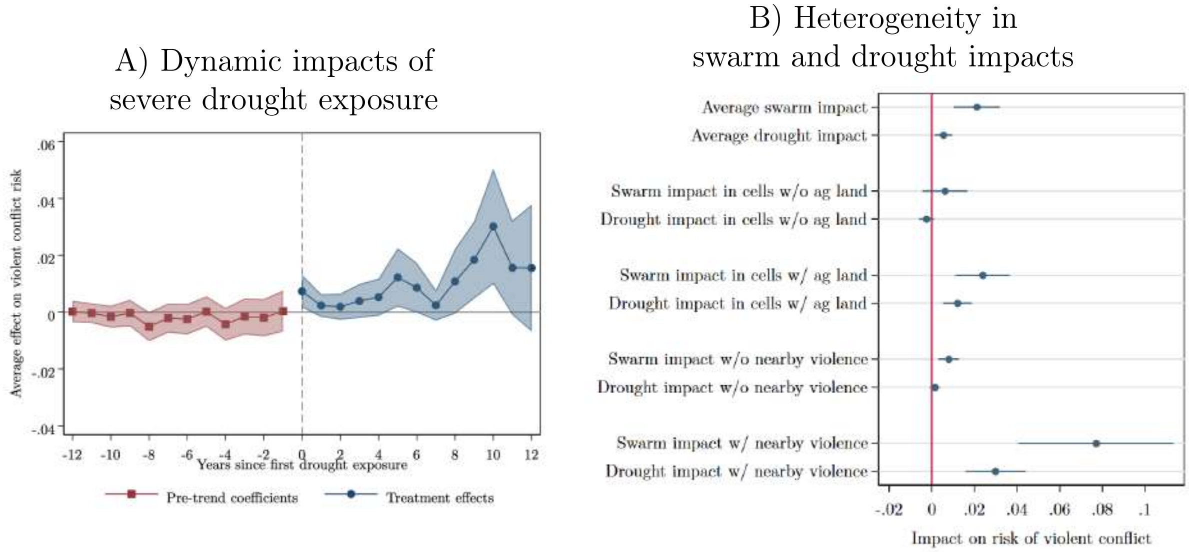
Locust swarms are a unique and catastrophic agricultural shock, but the model described in Section 3 is general and predicts similar patterns of impacts on the risk of violent conflict for other severe shocks to agricultural production. In this section I compare the impacts of locust swarm exposure to the impacts of exposure to a severe drought. Following Harari and La Ferrara (2018) and others I use the Standardized Precipitation and Evapotranspiration Index (SPEI) which combines both rainfall and the ability of the soil to retain water. Monthly grid cell SPEI data from 1996-2014 are included in the PRIO-GRID database (Begueria et al., 2014). The units for the measure of SPEI are standard deviations from the cell's historical average. I define a cell as experiencing a severe drought shock in a particular year if there are at least 4 consecutive months where the SPEI is below -1.5 (deviations within 1 indicate near normal conditions). As with locust swarm exposure I identify the first year in which a cell experiences such a drought and consider cells to be 'affected' in all subsequent years.

Figure 10 Panel A shows the results from an event study of drought exposure. Pre-exposure coefficients are small in magnitude and not statistically significant, with the exception of the period 8 years before exposure to a drought. Conflict risk increases by a statistically significant 0.9 percentage points in the year of exposure. Treatment effect estimates are positive and statistically significant for years 5-6 and 8-11 post-exposure, with the largest effects in the latter period. The average effect over the 12 years post-exposure is a 1.0pp increase in the annual risk of violent conflict.

Panel B shows TWFE estimates of the average long-term impacts of swarm and drought exposure by cell land cover and activity of armed groups in the surrounding cells. Impacts of locust swarms are the same as those reported previously. The TWFE estimate of the average impact of severe drought is a significant 0.5pp increase in the annual risk of violent conflict.

Although impacts of swarm exposure on violent conflict risk are larger, the patterns in the impacts of swarm and drought exposure are very similar consistent with both of these transitory agricultural shocks affecting the risk of violent conflict through the same mechanisms. Impacts of both shocks are concentrated in cells with any agricultural land, and in years where there is any violent conflict in surrounding cells. The difference in impact of past shock exposure in years where there is conflict activity near a cell is particularly large for locust swarms—a 6.8pp increase compared to 2.8pp for drought—likely because the sample for the analysis of swarm exposure includes more high-conflict years from 2015-2018 that are not included in the drought analysis sample. The similar patterns fore exposure to severe drought further indicate that the impacts of swarm exposure on violent conflict are not driven by bias in where locust swarms are reported.

Figure 10: Long-term impacts of drought and locust swarm exposure on violent conflict risk



Note: The dependent variable is a dummy for any violent conflict event observed. Panel A shows the event study results for impacts of exposure to any drought (4 consecutive months with SPEI<-1.5) on violent conflict risk over time. The shaded areas represent 95% confidence intervals.
Panel B shows coefficients and 95% confidence intervals are from three separate TWFE regressions for the average impacts of exposure to locust swarms and severe drought, respectively: one with no interactions and the others interacting all right-hand side variables first with a dummy for any agricultural (pasture or crop) land and then with a dummy for any violent conflict in the 15 other cells in the broader 1 degree cell in which the cell is located. Coefficients and standard errors are calculated using Stata's *xlincom* command based on the sums of the coefficients for the baseline effect and the interaction term.
In both panels, cells are regarded as 'affected' by locust swarms or drought for all years starting from the first year in which the shock is recorded in the cell. All regressions include country-by-year and cell fixed effects. Observations are grid cells approximately 28×28km by year. SEs clustered at the sub-national region level are in parentheses. Regression results for Panel B are shown in Table A8.

This finding has implications for previous research on the short-term impacts of economic shocks on conflict which also use grid cell panel data.[21] This literature has overwhelmingly focused on the short-term and assumes effects of shocks are temporary. A common empirical approach is a distributed lag two-way fixed effects model which takes the form:

$$Conflict_{ict} = \alpha + \beta_1 Shock_{ic,t} + \beta_2 Shock_{ic,t-1} + \delta X_{ict} + \gamma_{ct} + \mu_i + \epsilon_{ict} \tag{2}$$

This follows the persistent effects model in Equation 1 with the exception that instead of the *Shock* variable representing a permanent treatment status over subsequent years, in this temporary effects model the years following a shock are regarded as unaffected except as captured by the one year lag. This lag allows for limited delays or persistence in impacts of the shock (Burke et al., 2015).

With cell fixed effects the short-term impacts in the temporary effects model are esti-

[21]See for example Fjelde (2015), Harari and La Ferrara (2018), McGuirk and Burke (2020), McGuirk and Nunn (2021), and Ubilava et al. (2022). These studies analyze the impact of various shocks on conflict in Africa, and vary in their choice of controls and in the size of grid cells they analyze but all use a similar econometric specification.

mated relative to conflict risk in other years in the same cell where a shock is not observed, including years after exposure to a shock. For shocks that cause persistent increases in conflict, this implies that the temporary effects estimate will be biased downward as a result of comparing conflict risk in the year a shock is observed against later years with no shock but higher conflict risk *caused* by the shock. I show that this is the case for locust swarms and severe drought, comparing estimates from the temporary effects approach to the event study estimates treating shock effects as permanent Table A9.

In the case of the locust swarms, the temporary effects model estimates a highly significant 1.4pp *decrease* in the probability of any violent conflict in the year of exposure relative to unaffected cells. I can reject that this is the same as the event study estimate of 0.0pp with high confidence (p=0.009), consistent with downward bias of the temporary effects estimate. In the case of severe drought, the temporary effects estimate is a non-significant 0.4pp increase in violent conflict risk the year of a drought, compared to a highly significant 0.9pp increase in the event study estimate. While I cannot reject that the estimates for impact of drought on violent conflict risk in the year of exposure are the same (p=0.198), the two approaches would yield very different conclusions with different policy implications.

The temporary effects estimate for locust swarms—a 1.4pp decrease in violent conflict the year locusts are observed—is very close to Torngren Wartin (2018)'s estimate of a 1.3pp decrease using a similar method.[22] He interprets the result as suggesting endogenous under-reporting of locust swarm presence correlated with violent conflict, and does not consider potential bias from ignoring long-term impacts of swarm exposure on conflict risk. The much larger event study estimate for the impact of swarms on conflict in the same year suggests the large negative estimate in the temporary effects regression can instead be attributed to downward bias from ignoring long-term impacts.

These results provide some evidence of a potential misspecification of studies analyzing short-term impacts on conflict of transitory economic shocks that are severe enough to decrease permanent income and reduce wealth and future productivity. Studies of such shocks using specifications similar to Equation 2 and ignoring possible long-term effects will generate downward-biased short-term impact estimates to the extent the shocks increase long-term conflict risk.

[22] Torngren Wartin (2018) uses a similar sample and the analysis is at the level of 0.5° and 0.1° cells with somewhat different controls for lagged locust presence and weather but employs the same general distributed lag specification with cell and country-by-year fixed effects. He considers locust swarms and bands together while I focus on more destructive swarms alone, and includes some African countries with very few locust swarm observations over time while excluding Arabian countries with extensive locust activity.

7 Discussion

Exposure to a locust swarm leads to a large and persistent increase in the risk of violent conflict. Heterogeneity by timing of swarm exposure and crop land cover as well as similar patterns for exposure to severe drought indicate that these effects are not driven by areas where locust swarms are reported having different trends in conflict risk even in the absence of exposure.

Despite the large average long-term effects of swarm exposure, I find small and non-significant impacts of locust swarms on violent conflict in the year of exposure. This contrasts with much of the literature on climate and conflict which focuses on short-term effects and generally finds significant increases in conflict risk, though Bazzi and Blattman (2014) finds that commodity price shocks primarily affect conflict incidence and not conflict onset.

One explanation for the difference is that the effect of weather shocks on violent conflict may primarily go through channels other than agricultural production. Increases in both rainfall and temperatures significantly increase conflict risk in cells both without and without any cropland (Figure 5), echoing previous work finding impacts of rainfall on conflict that cannot be explained by agricultural mechanisms (Bollfrass and Shaver, 2015; Sarsons, 2015). But some studies also find that growing season weather shocks have null or inconsistent effects on conflict in the same year, notably Crost et al. (2018) and Harari and La Ferrara (2018). These studies do find increases in conflict risk the year after a shock, which I also find for locust swarms, and suggest a general mechanism suppressing immediate impacts of a shock on violent conflict.

A non-significant immediate effect of a negative agricultural shock on conflict risk could be explained in the context of the model as the decrease in opportunity cost of fighting being offset by a decrease in the returns to predation when the value of output and wealth available to capture is smaller. Studies of shocks to agricultural prices have shown instances where the predation mechanism outweighs the opportunity cost mechanism: McGuirk and Burke (2020) and Ubilava et al. (2022) both document increases in violent conflict in cells producing agricultural goods following increases in the global price of these goods.

Such an explanation would emphasize the need to consider both the opportunity cost and predation mechanisms in analyzing impacts of agricultural production shocks. The literature on production shocks and conflict has primarily emphasized the opportunity cost mechanism, but predictions based on this mechanism are not strongly supported by this study. In the short-term, an opportunity cost mechanism alone would predict larger immediate effects on the risk of violent conflict in cells exposed to a locust swarm. This mechanism would also predict larger impacts of past exposure to a shock in periods of subsequent agricultural production shocks which would further reduce productivity and opportunity costs of fighting, but I find no significant difference in impacts of swarm exposure in years with a severe drought. In addition, I would expect larger effects on factor conflict than output conflict if the null short-term effect is caused by offsetting decreases in the opportunity cost of fighting and the returns to capture of output, but I also find null short-term effects on factor conflict despite returns to factors not being affected by locust exposure.

In the long-term, an opportunity cost mechanism would imply a permanent income effect of locust swarm exposure which should be observable in measures of economic activity. I find a significant decrease in a measure of gross cell product or total income suggesting reduced economic activity. This aligns with evidence of decreased agricultural production and increased out-migration in exposed cells. But these effects are relatively small in magnitude and I find no effects on NDVI or average crop yield. While this may be due to the estimates being intent to treat effects with a small share of cells actually affected by exposure to a swarm, these results do not clearly show that swarm exposure decreases permanent income and the opportunity cost of fighting.

On the other hand, the evidence for the predation mechanism is strong. Impacts are greater for a measure of conflict over output—violence, rioting, or looting targeting civilians—than for a measure of conflict over territory or factors of production—UCDP violent conflict involving at least one organized actor and causing at least 25 deaths in a year. This difference aligns with expectations under a predation mechanism of individuals choosing to engage in fighting based on the returns they can expect to earn from this 'occupation.'

More strikingly, I find clear evidence of the importance of context and the conflict environment in particular for the impacts of a shock. Being exposed to a locust swarm makes cells significantly more likely to experience violent conflict when it emerges—and the feasibility of fighting and expected returns are higher and costs are lower—even years after exposure, but has limited effects otherwise and does not in general cause the onset of new violent conflicts. I find similar patterns for exposure to a severe drought. This is not a novel insight but has not been emphasized in the climate-conflict literature, and has important implications for considering what areas are most likely to become engaged in future conflicts triggered by various proximate causes.

The limited evidence in favor of the opportunity cost mechanism implies other mechanisms are playing a role in driving impacts of locust swarms on conflict risk. One possibility that is also income-related is an increase in local inequality, which may create discontent and cause conflict (Gurr, 2015). Several empirical studies show evidence of that greater inequality—particularly horizontal inequality between groups—is associated with increased conflict (Bartusevičius, 2014; Østby, 2008, 2013), though causal identification appears largely limited to panel analyses. Desert locust swarms are localized natural disasters and the median swarm would only cover 6% of the 0.25 degree cells that are the units of observation for this analysis. Swarm exposure would therefore have concentrated effects on agricultural production in only part of each cell, increasing within-cell inequality. Even if affected areas recover over time, they would grow from a lower base meaning the increased inequality would be likely to persist. Future work could test this empirically by analyzing outcomes for households over time at different distances from a locust swarm report. Another test could be to compare impacts of swarm exposure on violent conflict in cells with different levels of ethnic diversity, as past work indicates inequality-related grievances increase conflict more where the inequality is between rather than within groups.

Another potential mechanism is migration. Temporary migration following an agricultural shock may be preferred to fighting. Leaving to search of work is a common response to locust crop destruction (Thomson and Miers, 2002) and over 8 million people were displaced across East Africa as a result of the 2019-2021 locust outbreak (The World Bank, 2020). I do find significant increases in average long-term out-migration following locust exposure. If much of this is an immediate response to the agricultural productivity shock this could help explain the null short-term effects on local conflict risk. This mechanism would predict conflict spillovers in migrant destinations due to increased competition over local output and resources (see e.g., McGuirk and Nunn (2021), and Burke et al. (2024) for a review of this mechanism). I find marginally significant increases in conflict risk in areas within 100km from a locust swarm during the 2003-2005 upsurge which could be explained by a migration mechanism, but limited evidence of spillovers from swarm exposure overall. It is also unclear how this mechanism would explain the dynamic impacts of swarm exposure on violent conflict. A delay in effects could result from displaced populations returning and being faced with the effects of the severe income shock, but other mechanisms are likely more important in driving the delayed long-term increase in conflict risk.

Psychological mechanisms may also play a role, and have been documented in other studies of the climate-conflict relationship (see Burke et al. (2024) for a review). In this context, effects on religiosity may be important. The dominant religion in the sample countries is Islam, where locusts are mentioned as both a punishment from Allah and as a sign of Judgment Day. If locust swarms increase religiosity this may affect the perceived returns to fighting by increasing social, emotional, and supernatural costs, suppressing immediate violent conflict. But if increased religiosity persists this could contribute to some of the divisions that led to the onset of various civil conflicts and Islamic terror movements in the sample countries, increasing long-term violent conflict risk. This mechanism could be tested empirically by analyzing differences in impacts of swarm exposure on conflict by a measure of religious identity across cells, and by considering how swarm exposure affects measures of religiosity.

8 Conclusion

Violent conflict and environmental shocks can have devastating consequences for economic and human development which are the subject of significant study even beyond the economics literature. This paper shows that exposure to a severe agricultural production shock—both desert locust swarms and drought—significantly increases long-term conflict risk. I proposed a permanent income effect from initial agricultural destruction and coping strategies reducing later productivity and opportunity costs of fighting as a potential income-related mechanism, but find limited evidence of this. Locust swarms do not increase conflict risk in the year of exposure when the opportunity cost of fighting should be lowest, and impacts on conflict risk are largest after a delay of several years. While swarm exposure reduces long-term engagement in agriculture and promotes out-migration, consistent with many studies showing long-term economic impacts of natural disasters, but it does not persistently affect measures of agricultural productivity. Further research on long-term impacts of transitory economic shocks on household measures of productivity (as in Marending and Tripodi, 2022), labor supply, food security, and wealth would help further explore the potential role of this mechanism.

I find that locust swarm exposure does not immediately increase the risk of violent conflict, and that for both locust swarms and drought impacts are concentrated in periods of heightened insecurity due to other proximate causes Many factors have contributed to the increased onset and incidence of violent conflict in the sample countries in the sample period, including the Arab Spring, several civil wars, separatist movements, and the spread of terrorist organizations. The results emphasize the importance of the conflict environment and the role of the predation mechanism. This mechanism predicts larger impacts of a production shock when the feasibility of fighting or the expected returns are higher or when the costs (financial, societal, emotional) are lower. Exposure to locust swarms does not appear to cause affected populations to mobilize into armed groups, but may make them either more vulnerable to later attacks or more willing to join groups that mobilize around particular causes. The data do not allow me to distinguish whether populations in cells exposed to locust swarms are instigators or targets of the violent conflict events I observe; both cases are likely relevant given within-cell variation in swarm exposure. I show large impacts on a measure of conflict over output, which would target populations in exposed cells, but future work could attempt to shed more light on the nature of the violent conflicts that emerge following agricultural production shocks.

The findings suggest additional future avenues of research in the literature on climate and conflict. I show that the methods typically used in this literature, which treat shocks as only affecting conflict risk in the short-term, can result in downward-biased estimates of short-term effects when the shocks have long-term impacts. Event study analyses of severe weather shocks such as droughts could show the extent and patterns of long-term impacts on conflict risk and how this affects estimates of short-term effects. Although not a focus of this paper, the lack of variation in the impacts of rainfall and temperature deviations on violent conflict risk by land cover cast further doubt on whether effects on agricultural production are the primary mechanism. The association between climate and conflict has been demonstrated in a wide variety of settings but the mechanisms remain unclear (Burke et al., 2024; Mach et al., 2020). A better understanding of the different mechanisms is essential to determining the appropriate policy responses. Analyses of long-term impacts of agricultural shocks on measures of local inequality and on psychological factors could be particularly helpful in understanding both income- and non-income-related mechanisms.

The economic and human costs of increased conflict risk following severe agricultural shocks highlights the importance of policy efforts to respond to such shocks. Short-term efforts to provide food aid and other support to affected populations may be insufficient to facilitate full recovery and prevent future conflict risk. Burke et al. (2024) find robust evidence across studies that higher living standards reduce sensitivity of conflict risk to climate shocks. Additional research could explore whether policies that can promote resilience to agricultural shocks, such as cash transfers (Crost et al., 2016; de Janvry et al., 2006; Garg et al., 2020), livelihood graduation programs (Hirvonen et al., 2023), improved infrastructure (Gatti et al., 2021), and work programs (Fetzer, 2020) also reduce conflict risk.

The results also have implications for estimates of the economic and social costs of desert locust outbreaks. In addition to short-term impacts on agricultural production, incomes, and food security, previous studies have found that children exposed to locust swarms have worse education and health outcomes (Conte et al., 2023; De Vreyer et al., 2015; Le and Nguyen, 2022; Linnros, 2017). This study shows that swarm exposure also increases the long-term risk of conflict, which imposes further costs.

Past research on desert locusts has argued that limited impacts of outbreaks on aggregate national measures of agricultural production may mean expensive locust monitoring and control operations have limited net economic benefits (Joffe, 2001; Krall and Herok, 1997), though others have argued that local damages are extensive and motivate continued proactive locust control efforts (Showler, 2019; Zhang et al., 2019). Decisions about proactive locust control versus potential alternative responses—for example, increasing adoption of agricultural insurance or relying entirely on disbursement of relief after locust damages (Hardeweg, 2001)—should take the broader economic and social impacts of agricultural destruction by locusts into consideration. These costs should also motivate increased cross-country communication and collaboration in response to threats of locust swarms, particularly in light of the recent 2019-2021 outbreak.

Beyond contributing to our understanding of the relationship between agricultural production shocks and conflict risk, the findings are also relevant for considering multilateral policy around climate change mitigation and adaptation. Climate change is increasing the frequency and severity of agricultural shocks, including by creating conditions suitable for desert locust swarm formation. These shocks impose additional costs on society through their impacts on conflict risk which should be considered when weighing the costs and benefits of potential actions to reduce and respond to risks from agricultural shocks.

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9 Appendices

Appendix A: Additional Figures and Tables

Table A1: Summary statistics

Panel A: Yearly variables

	Mean	SD	Min	50^{th}	75^{th}	Max	N
Any violent conflict event - ACLED	0.02	0.13	0.0	0.0	0.0	1.0	561726
Any protest or riot event - ACLED	0.01	0.09	0.0	0.0	0.0	1.0	561726
Any violent conflict event - UCDP	0.01	0.10	0.0	0.0	0.0	1.0	561726
Any swarms in cell	0.00	0.07	0.0	0.0	0.0	1.0	561726
Any swarms within 100km outside cell	0.05	0.21	0.0	0.0	0.0	1.0	561726
Any swarms within 100-250km of cell	0.11	0.32	0.0	0.0	0.0	1.0	561726
Population (10,000s)	1.63	8.91	0.0	0.1	0.9	749.8	558272
Total annual rainfall (100 mm)	2.40	3.79	0.0	0.8	2.8	43.4	558184
Max annual temperature (deg C)	37.55	5.10	11.5	38.2	41.3	49.1	558184

Panel B: Fixed variables

	Mean	SD	Min	50^{th}	75^{th}	Max	N
Any ACLED violent conflict event in cell from 1997-2018	0.13	0.33	0.0	0.0	0.0	1.0	25533
Any protest/riot event in cell from 1997-2018	0.07	0.25	0.0	0.0	0.0	1.0	25533
Any UCDP violent conflict event in cell from 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25533
Any locust swarm in cell from 1985-2023	0.12	0.33	0.0	0.0	0.0	1.0	25533
Any locust swarm in cell from 1997-2018	0.09	0.29	0.0	0.0	0.0	1.0	25533
Any locust swarm within 100km from 1997-2018	0.55	0.50	0.0	1.0	1.0	1.0	25533
First exposed to locust swarm between 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25533
First exposed to locust swarm in 2003-2005 upsurge	0.05	0.22	0.0	0.0	0.0	1.0	25533
Any cropland or pasture in cell	0.57	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to crops or pasture	0.23	0.32	0.0	0.0	0.4	1.0	25435
Any pasture in cell	0.56	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to pasture	0.18	0.27	0.0	0.0	0.3	1.0	25435
Any cropland in cell	0.31	0.46	0.0	0.0	1.0	1.0	25435
Share of cell allocated to crops	0.05	0.13	0.0	0.0	0.0	1.0	25435

Note: Observations are grid cells approximately 28×28km by year. Values for land cover are for the year 2000.

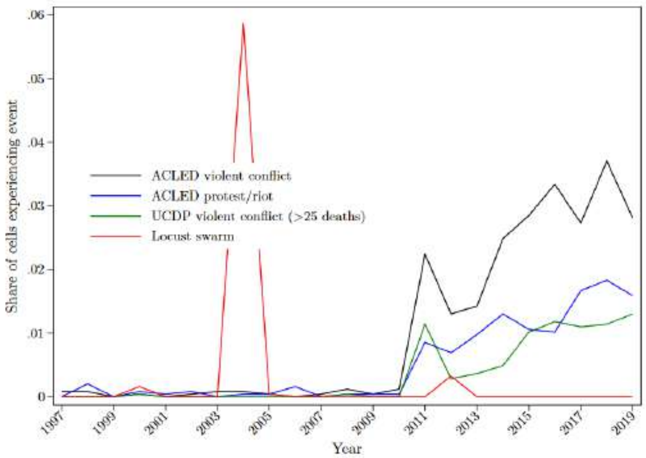
Table A2: Balance in cell characteristics by exposure to any locust swarm

	All cells		W/in 100km of any swarm		All cells, exposure IPW	
	Control Mean (SD)	Treat Diff. (SE)	Control Mean (SD)	Treat Diff. (SE)	Control Mean (SD)	Treat Diff. (SE)
Population in 2000 (10,000s)	1.22	1.32***	1.56	0.98***	1.23	-0.00
	[6.94]	(0.37)	[8.35]	(0.33)	[6.97]	(0.43)
Mean of cell nightlights	0.05	0.01**	0.05	0.00	0.05	-0.00
1996-2012 (0-1)	[0.04]	(0.00)	[0.04]	(0.00)	[0.04]	(0.00)
Distance to national capital	707.75	-182.42***	611.63	-86.30**	707.37	91.46
(km)	[406.53]	(45.59)	[372.80]	(34.71)	[406.67]	(74.08)
Percent of cell covered by	4.66	0.85	5.50	0.01	4.70	0.03
crop land in 2000	[13.06]	(0.77)	[14.05]	(0.68)	[13.12]	(1.50)
Percent of cell covered by	17.58	10.86***	21.17	7.27***	17.73	-5.23*
pasture land in 2000	[26.75]	(2.38)	[27.96]	(2.20)	[26.83]	(3.01)
Percent of cell covered by	68.89	-9.93***	62.50	-3.55	68.61	6.59
barren area in 2009	[42.65]	(3.64)	[43.23]	(3.11)	[42.75]	(6.49)
Percent of cell covered by	0.08	0.10	0.09	0.09	0.08	-0.02
urban area in 2009	[0.75]	(0.07)	[0.85]	(0.07)	[0.75]	(0.03)
Mean annual rainfall 1997-2018	2.40	0.16	2.57	-0.01	2.42	-0.44
(100 mm)	[3.81]	(0.24)	[3.72]	(0.17)	[3.82]	(0.39)
Mean annual max temperature	37.66	-1.65***	36.92	-0.91**	37.64	0.45
1997-2018 (deg C)	[5.11]	(0.45)	[5.06]	(0.38)	[5.12]	(0.73)
Mean of cell annual share of	0.09	-0.00	0.09	-0.00	0.09	-0.01*
months with drought 1998-2014	[0.03]	(0.00)	[0.04]	(0.00)	[0.03]	(0.00)
		$F = 7.06$		$F = 3.35$		$F = 1.56$
Joint significance		$p < 0.01$		$p < 0.01$		$p = 0.03$

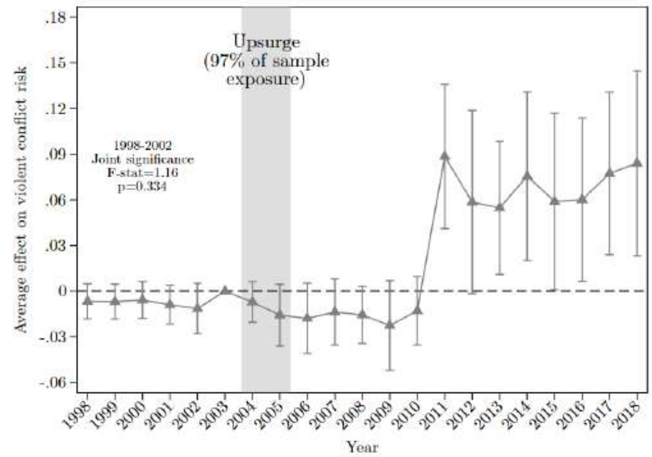
Note: The table shows results from separate bivariate regressions of baseline or mean cell characteristics on a dummy for being exposed to a locust swarm during the study period. The rows indicate which dependent variable is used. The first set of columns includes all cells while the second restricts the sample to cells within 100km of any locust swarm observation from 1996-2021. The third set of columns includes all cells but weights observations by the inverse of the estimated propensity to have been exposed to a locust swarm during the study period. I include results of joint tests that there is no relationship between any of the characteristics and swam exposure. Observations are grid cells approximately 28×28km by year. SEs clustered at the sub-national region level are in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A1: Changes in conflict environment and locust exposure event studies by country

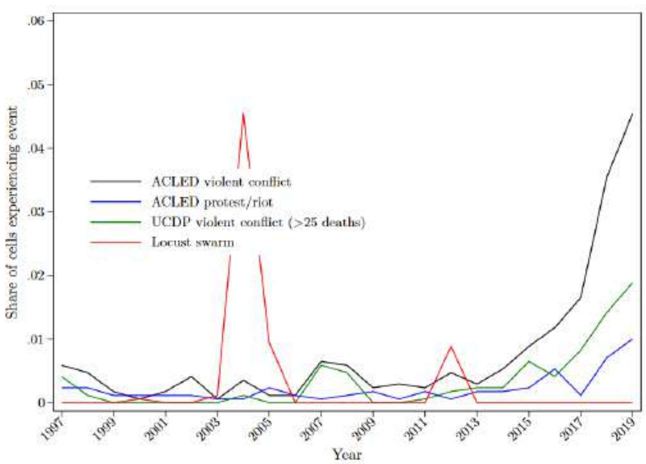
A) Libya swarm and conflict trends



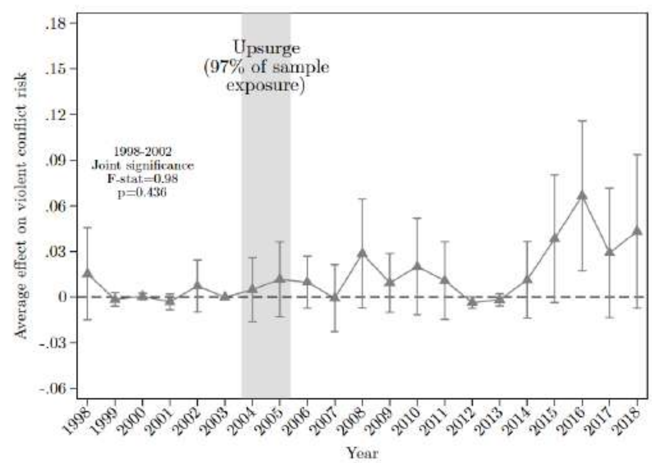
B) Libya upsurge event study



C) Niger swarm and conflict trends



D) Niger upsurge event study



Note: The figure replicates Figure 4 for Libya and Niger alone.

Table A3: Average impacts of exposure to locust swarms on violent conflict risk by land cover

Outcome: Any violent conflict event	(1) All land	(2) Land = Any crop or pasture land	(3) Land = Any crop land
Exposed to any locust swarm	0.020***	0.005	0.005
	(0.005)	(0.006)	(0.005)
Total annual precipitation (SDs)	0.004**	0.001	0.002*
	(0.002)	(0.001)	(0.001)
Max annual temperature (SDs)	0.019***	0.016***	0.017***
	(0.006)	(0.006)	(0.006)
Population (10,000s)	0.009***	0.008***	0.007
	(0.002)	(0.002)	(0.006)
Exposed to any locust swarm=1 × Land=1		0.017*	0.022**
		(0.009)	(0.008)
Total annual precipitation (SDs) × Land=1		0.005**	0.003
		(0.002)	(0.002)
Max annual temperature (SDs) × Land=1		0.004	0.003
		(0.005)	(0.004)
Population (10,000s) × Land=1		0.001	0.001
		(0.003)	(0.006)
Observations	327646	327646	327646
Outcome mean post-2004, no exposure	0.028	0.028	0.028
Country-Year FE	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Note: The table presents results shown in Figure 5. The 'Land=1' rows show the coefficients for the interaction of right-hand side variables with cell land cover dummies indicated in the column heading. The outcome mean for control cells is shown for post-2004 for comparison with exposure impacts in the period after the majority of swarm exposure occurred.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Impacts of swarm exposure by land cover and swarm timing

Outcome: Any violent conflict event	(1)
Off/planting season Non-crop cell	0.001 (0.006)
Off/planting season Crop cell	0.008 (0.007)
Growing/harvest season Non-crop cell	0.011 (0.007)
Growing/harvest season Crop cell	0.027*** (0.008)
Observations	327646
Outcome mean post-2004, no exposure	0.028
p, off/plant season non-crop=crop effect	0.368
p, grow/harvest season non-crop=crop effect	0.131
p, non-crop off/plant=grow/harvest effect	0.311
p, crop off/plant=grow/harvest effect	0.087
Country-Year FE	Yes
Cell FE	Yes
Controls	Yes

Note: The table presents results shown in Figure 6.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Average impacts of locust swarm exposure on different conflict types

	N	Never exposed Mean (SD)	Swarm exposure treatment (SE)
Any violent conflict event - ACLED	327646	0.021	0.020***
(Battles, explosions, or violence against civilians)		(0.143)	(0.005)
Any violent state conflict - ACLED	327646	0.013	0.014***
(At least one state actor)		(0.113)	(0.003)
Any violent non-state conflict - ACLED	327646	0.014	0.014***
(No state actor)		(0.118)	(0.004)
Any violent one-sided conflict - ACLED	327646	0.012	0.014***
(Any civilian engagement)		(0.108)	(0.004)
Any conflict targeting civilians - ACLED	327646	0.013	0.017***
(Violence against civilians, riots, or looting)		(0.115)	(0.004)
Any protest or riot event - ACLED	327646	0.009	0.022***
(Protests or riots)		(0.094)	(0.004)
Any violent conflict event - UCDP	327646	0.010	0.007**
(At least one organized actor and >25 deaths in year)		(0.102)	(0.003)
Any violent state conflict - UCDP	327646	0.008	0.005*
(At least one state actor)		(0.086)	(0.003)
Any violent non-state conflict - UCDP	327646	0.002	0.002*
(No state actor)		(0.043)	(0.001)
Any violent one-sided conflict - UCDP	327646	0.002	0.002
(Any civilian engagement)		(0.050)	(0.001)
Total fatalities - ACLED	327646	0.735	0.882
		(65.031)	(0.611)
Total fatalities - UCDP	327646	0.594	0.645
		(107.449)	(0.587)

Note: Each row replicates Table A3 column 1 for a different conflict outcome. The dependent variables are dummies for any conflict event being observed in a cell in a year, with the conflict type specified in each row. See the figure note for Table A3 for more detail.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Heterogeneity in impacts of exposure to locust swarms on violent conflict risk by intensity of conflict environment

Outcome: Any violent conflict event	(1) Any violent conflict in surrounding 1 deg cell	(2) Any violent conflict in surrounding sub-region	(3) Post-onset of Arab Spring in country	(4) Post-onset of civil conflict in country	(5) Post-onset of separatist movement in country	(6) Post-onset of Islamic terror groups in country	(7) Any active famine in country	(8) Any active drought in cell
Exposed to any locust swarm	0.007***	0.006**	0.009***	0.002	0.006*	0.006	0.017***	0.011***
	(0.002)	(0.002)	(0.003)	(0.004)	(0.003)	(0.006)	(0.005)	(0.003)
Conflict environment=1	0.119**	0.091***						0.021*
	(0.050)	(0.035)						(0.012)
Exposed to any locust swarm	0.062***	0.028***	0.036***	0.032***	0.040***	0.024*	0.028**	0.002
× Conflict environment=1	(0.014)	(0.007)	(0.011)	(0.012)	(0.015)	(0.013)	(0.014)	(0.003)
Observations	327646	327646	327646	327646	327646	327646	327646	267644
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

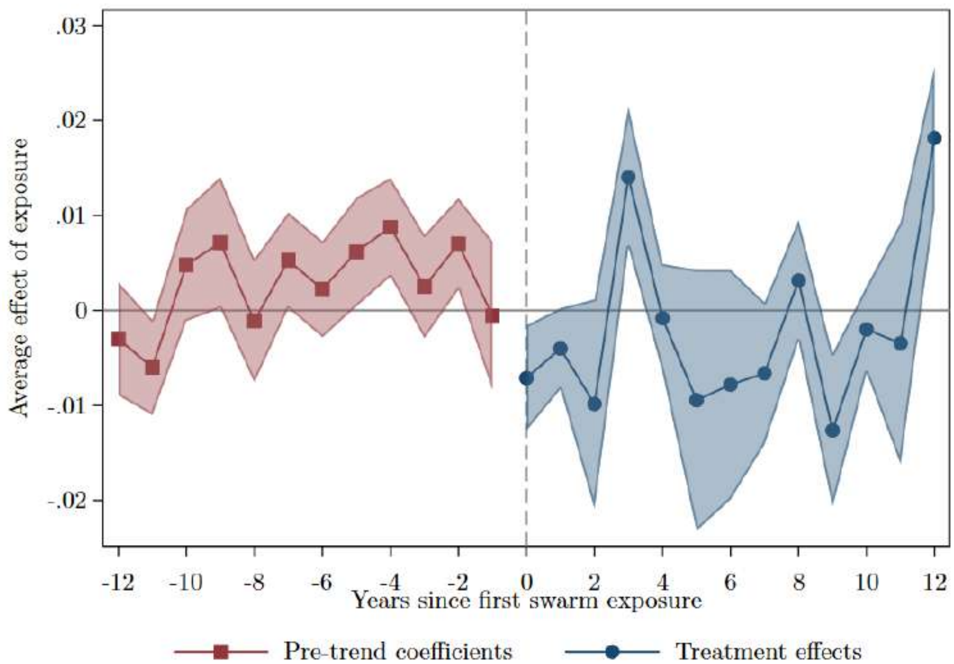
Note: The table presents results shown in Figure 8. The conflict environment variables in columns 3-6 are defined at the country level based on the timing that Arab Spring uprisings, civil conflicts, separatist movements, or Islamic terror attacks began in the country, with all subsequent years coded as under this conflict environment. Countries not exposed to such conflicts are coded as 0 for all years. Column 7 is also defined at the country level but varies by year rather than being 'turned on' after an initial famine exposure in the sample period.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Average impacts of locust swarm exposure on measures of productivity

	(1) Max annual NDVI (all cells)	(2) Gross cell product, 1990 USD PPP millions	(3) Net annual cell migration per 1000 people	(4) Mean cluster crop cultivated area (ha)	(5) Mean cluster crop total production (metric tons)	(6) Mean cluster crop average yield (kg/ha)	(7) Mean cluster TLU per sq. km
Exposed to any locust swarm	0.002	-8.085**	-12.989**	-119.314**	-868.942**	-55.981	-1.530
	(0.001)	(3.496)	(5.320)	(53.351)	(402.078)	(122.219)	(3.023)
Total annual rainfall (100 mm)	0.006***	5.433**	-0.989	15.146	147.640*	53.825	2.679
	(0.001)	(2.364)	(0.696)	(9.255)	(75.629)	(32.528)	(3.112)
Total annual rainfall prior year (100 mm)	0.002***	0.882	-0.900	21.666	41.017	-7.909	-0.071
	(0.000)	(1.018)	(0.633)	(13.974)	(77.093)	(14.914)	(0.827)
Max annual temperature (deg C)	-0.002***	-6.416**	-3.165**	-56.137	-512.576	-152.554	-1.150
	(0.001)	(2.750)	(1.521)	(51.270)	(742.876)	(101.725)	(3.632)
Max annual temperature prior year (deg C)	0.003***	4.866*	-2.550*	5.844	-230.108	-87.870	-7.742*
	(0.001)	(2.633)	(1.510)	(56.337)	(864.585)	(159.075)	(3.918)
Population (10,000s)	0.000	9.548***	25.861*	1.765	8.877	-2.417	0.072
	(0.000)	(2.266)	(14.367)	(1.125)	(31.024)	(2.594)	(0.089)
Observations	466569	42350	454176	4256	4256	4171	4256
Outcome mean, no swarms	0.225	59.16	-0.202	2602.8	15740.6	3868.7	42.75
Proportional effect of swarm	0.008	-0.137	64.181	-0.046	-0.055	-0.014	-0.036
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table presents results from the outcomes shown in Figure 9, but without normalizing the outcome variables. NDVI 1 is constructed from MODIS satellite data. I take the maximum of annual 16-day NDVI observations in each cell. Gross cell product is an estimate of cell-level total income for 2000 and 2005 from Nordhaus (2006) as included in the PRIO-GRID database. Net annual cell-level migration is from Niva et al. (2023). Outcome variables in columns 4-7 from the DHS AReNA database (IFPRI 2020) at the level of the clusters where DHS surveys are conducted, and are measured only in survey years. 'TLU' indicates Tropical Livestock Units. All regressions include country-by-year and location fixed effects. SEs are clustered at the sub-national region level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A2: Impacts of exposure to locust swarms on max annual NDVI over time, crop cells



Note: NDVI is calculated from MODIS satellite data, and I take the maximum of annual 16-day NDVI observations in each cell-year. The event study is restricted to crop cells, where NDVI is a potential proxy for agricultural production. Observations are grid cells approximately 28×28km by year.

Table A8: Average impacts of exposure to agricultural shocks on violent conflict risk

Outcome: Any violent conflict event	(1) Swarm	(2) Swarm	(3) Swarm	(4) Drought	(5) Drought	(6) Drought
Exposed to shock	0.020***	0.005	0.007***	0.005**	-0.003	0.001
	(0.005)	(0.005)	(0.002)	(0.002)	(0.002)	(0.001)
Exposed to shock × Any cropland or pasture in cell		0.018**			0.015***	
		(0.007)			(0.004)	
Any violent conflict elsewhere in 1 degree cell			0.123**			0.053*
			(0.061)			(0.031)
Exposed to shock × Any violent conflict elsewhere in 1 degree cell			0.068***			0.028***
			(0.016)			(0.007)
Observations	308637	304127	308637	432843	429524	432843

Note: The table presents results shown in Figure 10. The dependent variable is a dummy for any violent conflict event observed. The columns indicate which agricultural shock is analyzed. Cells are regarded as 'affected' by a shock for all years starting from the first year in which the shock is recorded in the cell. Columns with interactions also include interactions with other right-hand side variables. Observations are grid cells approximately 28×28km by year for 1997-2018 for swarms and 1997-2014 for drought. The sample for impacts of swarm exposure is restricted to cells within 100km of a swarm observation. SEs clustered at the sub-national region level are in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Impacts of agricultural shocks on the conflict risk, treating shocks as temporary vs. persistent

	(1)	(2)	(3)	(4)
	Locust swarm		Severe drought	
	Short-term	Long-term	Short-term	Long-term
Any shock in cell	-0.014***		0.004	
	(0.004)		(0.003)	
Any shock in cell previous year	-0.006		0.002	
	(0.004)		(0.002)	
Exposed to shock		0.020***		0.005**
		(0.005)		(0.002)
Year 0 event study estimate	0.000		0.009***	
	(0.002)		(0.003)	
p-value, equality of estimates	0.009		0.198	
Observations	320184	294606	408744	408752
Year FE	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Note: The dependent variable is a dummy for any violent conflict event observed. Locust swarm and drought exposure in columns (1) and (3) is based only on whether the shock was observed in a particular year. Cells are considered 'affected' by any locust swarm or any drought in columns (2) and (4) for all years after these shocks are first observed in the cell. At the bottom of columns (1) and (3) I show coefficients for the treatment effect in the year of exposure from a persistent effects event study specification along with the p-value for the test of equality of the coefficients. Controls in all regressions include total cell population and current and prior year measures of total rainfall and maximum annual temperature. Observations are grid cells approximately 28×28km by year for 1997-2018 for swarms and 1997-2014 for drought. The sample for impacts of swarm exposure is restricted to cells within 100km of a swarm observation. SEs clustered at the sub-national region level are in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

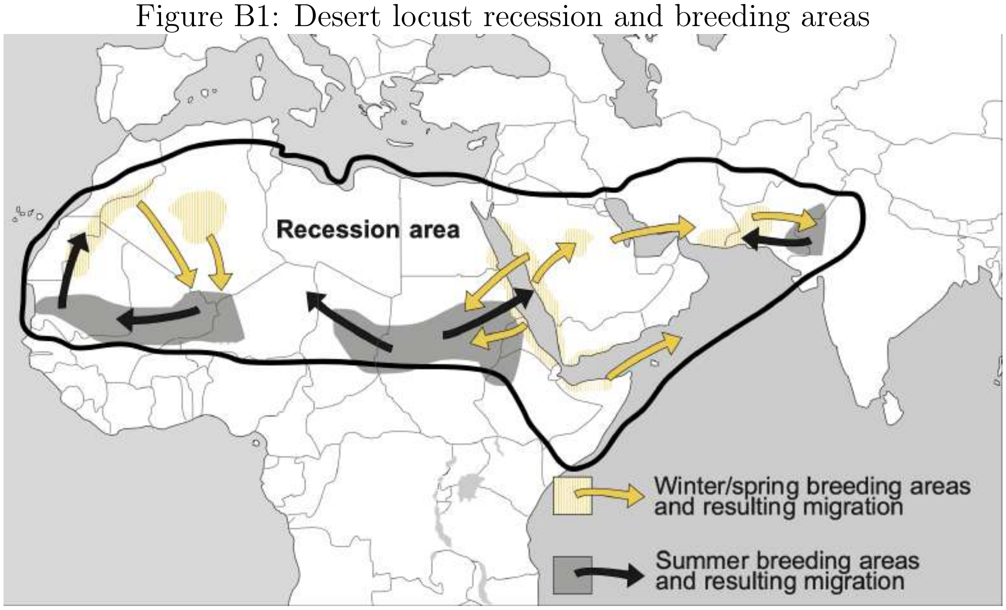
Appendix B: Desert locusts background

The desert locust is considered the world's most dangerous and destructive migratory pest (Cressman et al., 2016; Lazar et al., 2016). Locusts consume any available vegetation, and swarms frequently lead to the total destruction of local agricultural output (Showler, 2019). Damages from locust shocks can be extreme, with a small swarm covering one square kilometer can consume as much food in one day as 35,000 people. During the last locust upsurge in 2003-2005 in North and West Africa, 100, 90, and 85% losses on cereals, legumes, and pastures respectively were recorded, affecting more than 8 million people (Brader et al., 2006; Renier et al., 2015). Damages to crops alone were estimated at \$2.5 billion USD and \$450 million USD was required to bring an end to the upsurge (ASU, 2020).

In the most recent upsurge from 2019-2021 in East Africa and the Arabian Peninsula, over 40 million people in 10 countries faced severe food insecurity due to crop destruction. Locust control operations undertaken by the United Nations Food and Agriculture Organization (FAO) and its partners, primarily via ground and aerial spraying of pesticides, and global food aid efforts helped reduce the damages (Food and Agriculture Organization of the United Nations (FAO), 2022b). The FAO estimates that 3.5 million people were affected by locust destruction, but that control efforts saved agricultural production worth \$1.7 billion USD.

Small numbers of locusts are always present in desert 'recession' areas from Mauritania to India (Figure B1). The population can grow exponentially under favorable climate conditions: periods of repeated rainfall and vegetation growth overlapping with the breeding cycle. The 2019-2021 upsurge persisted in large part because of repeated heavy precipitation out of season due to cyclones, prompting explosive reproduction (Cressman and Ferrand, 2021). The 2003-2005 upsurge was initiated by good rainfall over the summer of 2003 across four separate breeding areas. This was followed by two days of unusually heavy rains in October 2003 from Senegal to Morocco, after which environmental conditions were favorable for reproduction over the following 6 months (FAO and WMO 2016).

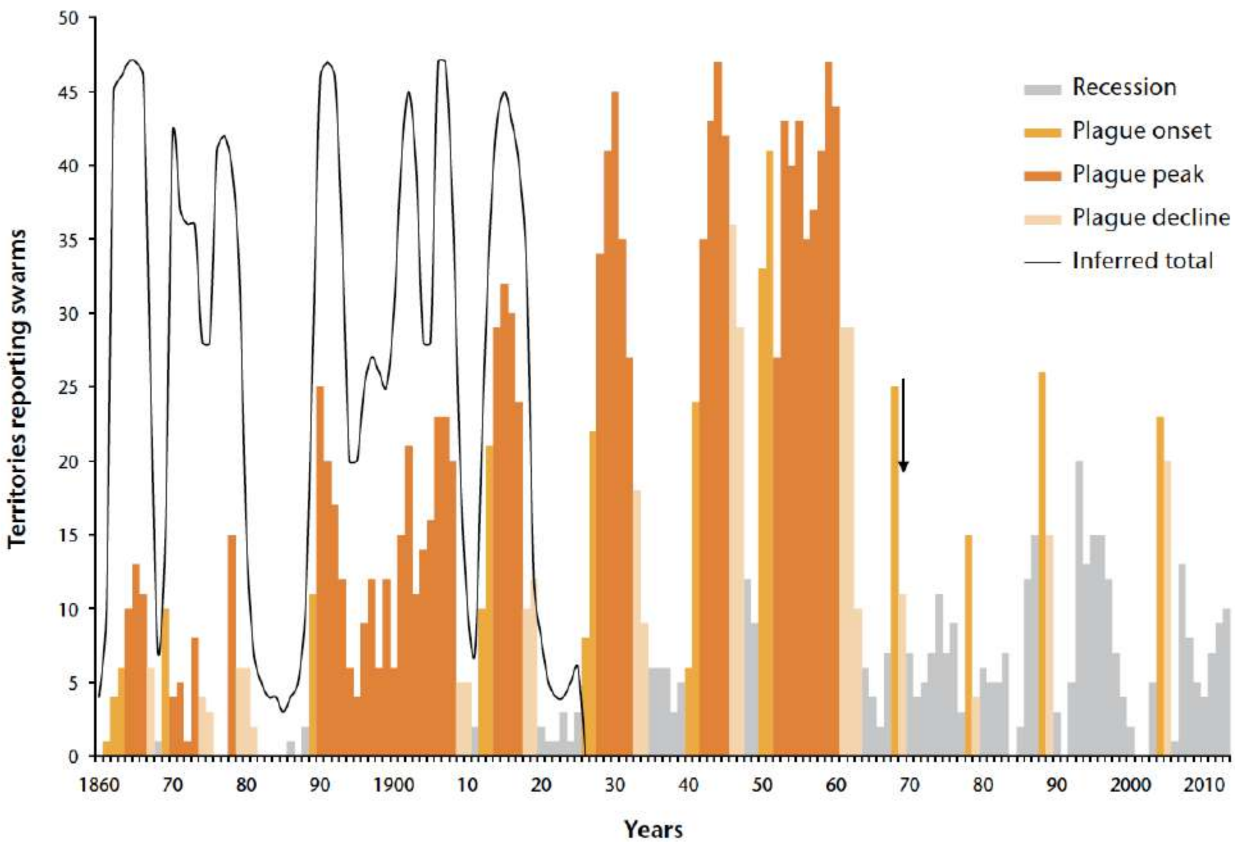
Figure B1: Desert locust recession and breeding areas



Source: Symmons and Cressman (2001).

Unique among grasshopper species, after reaching a particular population density desert locusts undergo a process of 'gregarization' wherein they mature physically and form large bands or swarms which move as a cohesive unit (Symmons and Cressman, 2001). Locust bands may extend over several kilometers and alternate between roosting and marching, typically downwind (FAO and WMO 2016). Locust swarms form when bands of locusts remain highly concentrated when they reach the adult stage and become able to fly. This formation of swarms can lead to 'outbreaks,' where locusts spread out from their largely desert initial breeding areas. Locusts in swarms have increased appetites and accelerated reproductive cycles, and are thus particularly threatening to agriculture. The FAO distinguishes different levels of locust swarm activity (Symmons and Cressman, 2001). I use the terms 'outbreak' and 'upsurge' interchangeably to refer to any locust swarm activity. By the FAO definition 'outbreaks' refer to localized increases in locust numbers while 'upsurges' refer to broader and more sustained locust activities. A third level, 'plagues,' is characterized by larger and more widespead locust infestations. Few locust swarms are observed outside of major outbreaks, as conditions favoring swarm formation tend to produce large swarms which reproduce and spread rapidly and are very difficult to control.

Figure B2: Desert locust observations by year

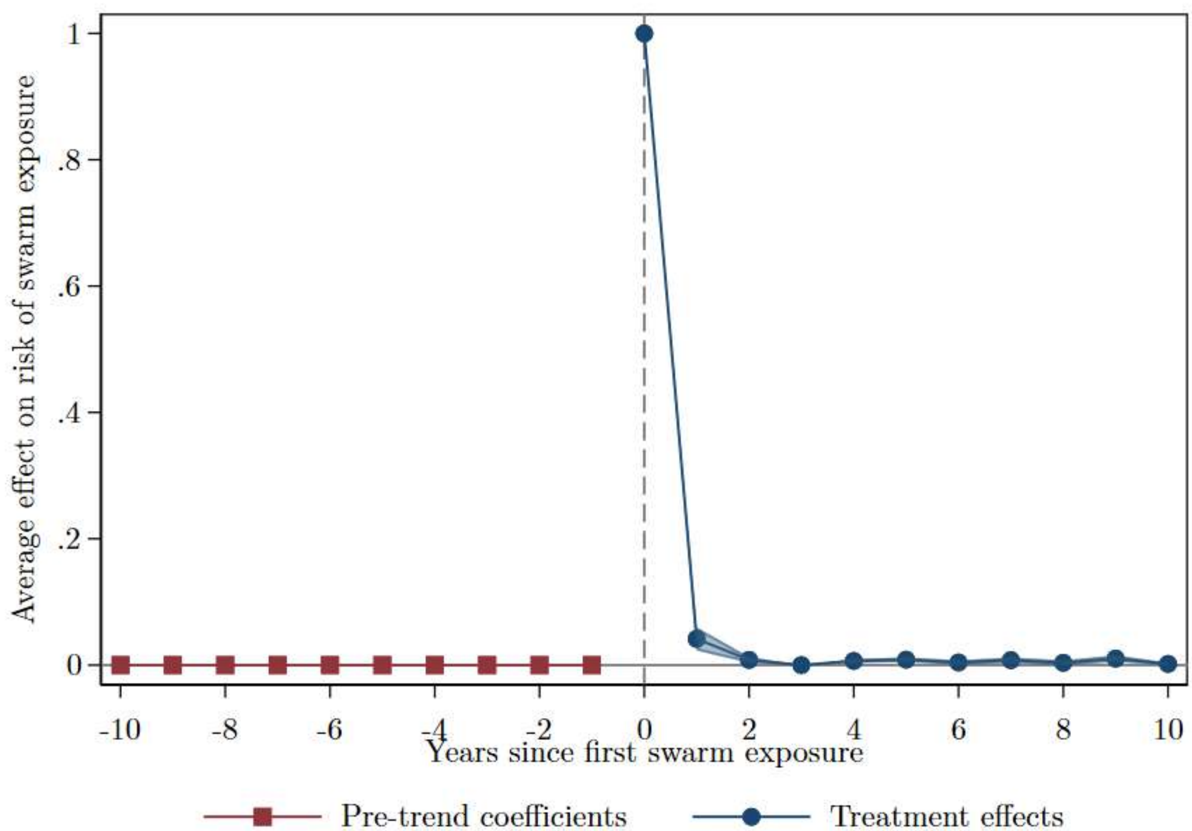


Source: Cressman and Stefanski (2016), Figure 6.

The frequency of large-scale outbreaks has fallen since around the 1980s (Figure B2), in large part due to increases in coordinated preventive operations (Cressman and Stefanski, 2016). Given their tolerance for extreme heat and responsiveness to periods of heavy precipitation, however, climate change might create conditions conducive to more frequent desert locust outbreaks.

As illustrated by Figure 1, locust swarms are not observed with any regularity over time or space. Desert locusts are migratory, moving on after consuming all available vegetation, and outside of outbreak periods are ultimately restricted to desert 'recession' areas. Unlike many other insect species, therefore, the arrival of a desert locust swarm does not signal a permanent change in local agricultural pest risk. Instead, the arrival of a swarm can be considered a locally and temporally concentrated natural disaster where all crops and pastureland are at risk (Hardeweg, 2001). Figure B3 illustrates how exposure to a locust swarm does not significantly affect the risk of exposure over the following years, consistent with exposure being a function of quasi-random variations in wind patterns and flight duration during swarm outbreaks.

Figure B3: Impact of swarm exposure on future exposure risk



Note: By construction, none of the exposed areas had any locust swarm recorded in the years preceding their first exposure since 1990.

Locust swarms vary in their density and extent (Symmons and Cressman, 2001). The average swarm includes around 50 locusts per m^2 with a range from 20-150, and can cover under 1 square kilometers to several hundred (Symmons and Cressman, 2001). About half of swarms exceed 50km^2 in size (FAO and WMO 2016), meaning swarms typically include over a billion individuals. Swarms fly downwind from a few hours after sunrise to an hour or so before sunset when they land and feed. Swarms do not always fly with prevailing winds and may wait for warmer winds. Small deviations in the positions of individual locusts in the swarm can also lead to changes in swarm flight trajectory, making their movements difficult to predict. Seasonal changes in these winds tend to bring locusts to seasonal breeding areas at times when rain and the presence of vegetation is most likely, allowing them to continue breeding (FAO and WMO 2016).

These movement characteristics inform efforts to predict locust swarm movements, but these remain highly imprecise. The desert locust bulletins produced monthly by the FAO include forecasts of areas at risk of desert locust activity, but the areas described are quite large, often encompassing several countries in periods with increased swarms. While breeding regions and the broad areas at risk over different time periods can generally be predicted with some accuracy (Latchininsky, 2013; Samil et al., 2020; Zhang et al., 2019), predicting specific local variation in swarm presence remains a challenge due to the multiple factors influencing specific flight patterns (FAO and WMO 2016).

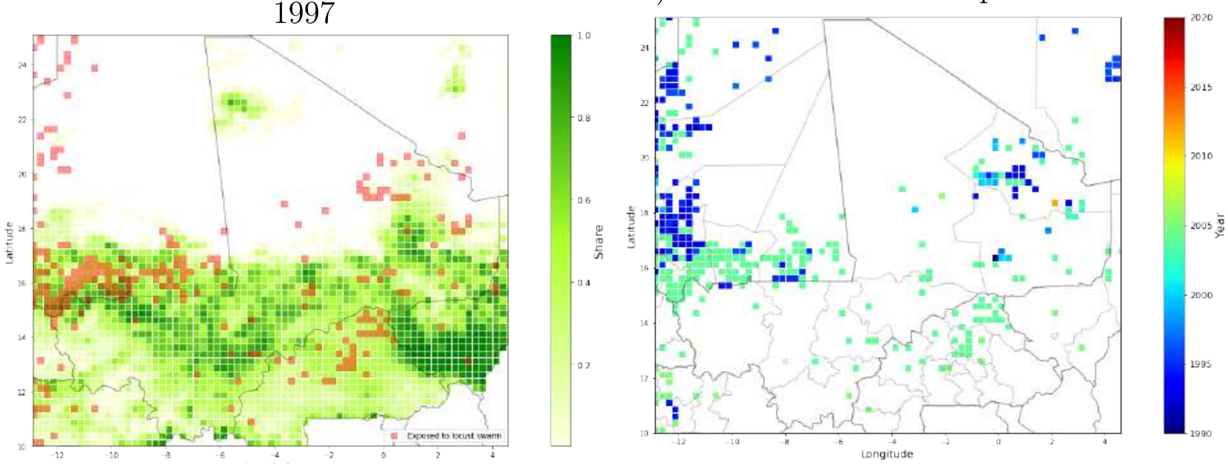
Patterns in swarm movements lead to local variation in locust swarm exposure. After taking off, swarms fly for 9-10 hours rather than landing as soon as they encounter new vegetation. A swarm can easily move 100km or more in a day even with minimal wind (Symmons and Cressman, 2001). Consequently, the flight path of a locust swarm will include both affected and unaffected areas, with the affected areas determined by largely by patterns of wind direction and speed over time from the initial swarm formation in breeding areas.

Figure B4 illustrates the variation in areas affected by locust swarms over space around Mali. Swarm reports are densely clustered in the breeding areas in southern Mauritania where locust swarms reproduced in summer 2004. Outside of this area there is considerable variation in where swarms were reported, with distances between reported swarms over time consistent with typical flight distances.

Figure B4: Reports of locust swarms around Mali

A) Areas exposed to locust swarms since 1997

B) Year of first swarm exposure since 1990



Note: The figure illustrates the grid cells exposed to locusts swarms for the area around the country of Mali. Locust swarm reports are from the FAO Locust Watch database. Panel A overlays these reports on a map of the share of agricultural land area in each cell, while Panel B illustrates the timing of first exposure to locust swarms.

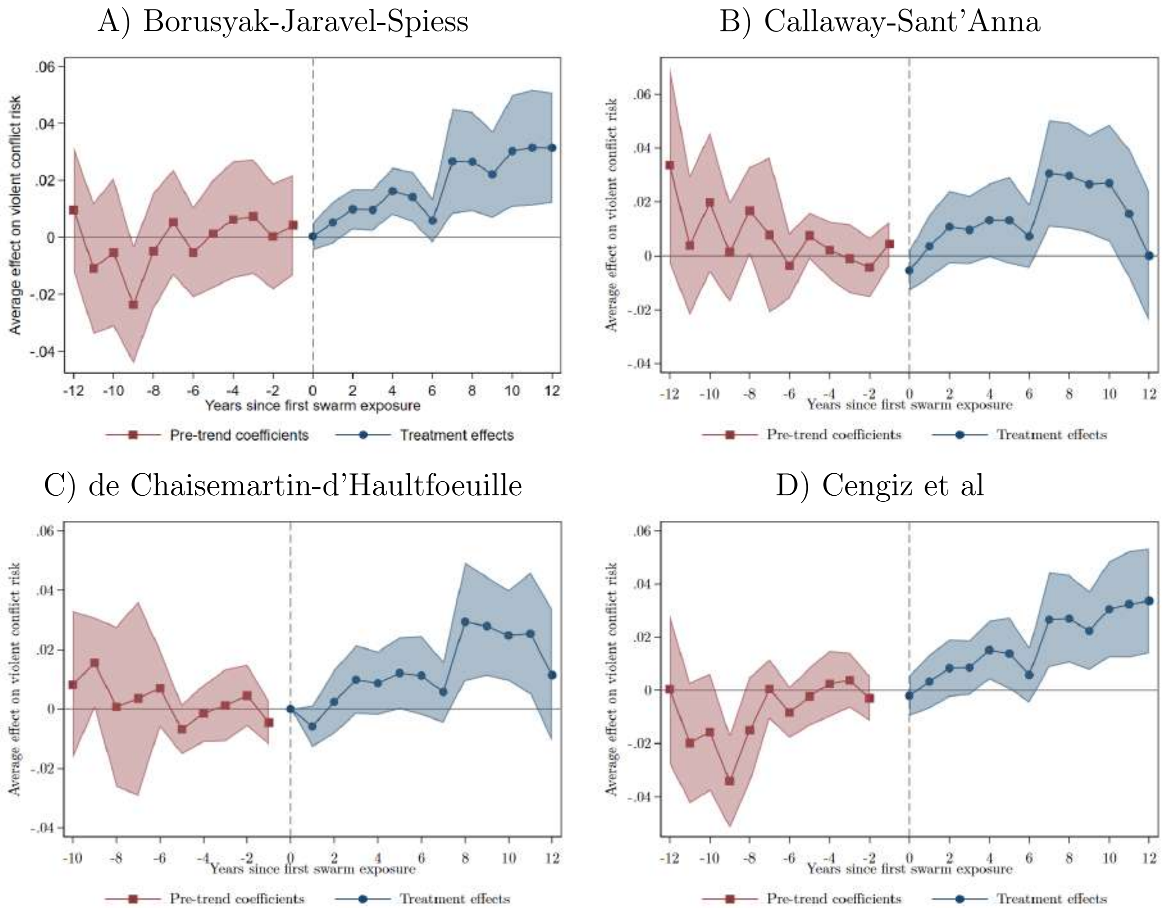
An important result of the local variation in locust swarm damages during outbreaks is that macro level impacts may be muted, since outbreaks occur in periods of positive rainfall shocks which tend to increase agricultural production in unaffected areas. Several studies find that impacts of locust outbreaks on national agricultural output and on prices are minimal, despite devastating losses in affected areas (Joffe, 2001; Krall and Herok, 1997; Showler, 2019; Thomson and Miers, 2002; Zhang et al., 2019). Chatterjee (2022) finds that wheat yields are 12% lower on average in Indian districts typically affected by desert locusts in years of locust outbreaks, in contrast to very large decreases in the specific areas exposed to locust swarms in those years.

Farmers have no proven effective recourse when faced with the arrival of a locust swarm, though activities such as setting fires, placing nets on crops, and making noise are commonly attempted. While these may slow damage they have little effect on locust population or total damages (Dobson, 2001; Hardeweg, 2001; Thomson and Miers, 2002). Locust outbreaks end due to a combination of migration to unfavorable habitats, failure of seasonal rains in breeding areas, and control operations (Symmons and Cressman, 2001). The only current viable method of swarm control is direct air or ground spraying with pesticides (Cressman and Ferrand, 2021). These control operations do not prevent immediate agricultural destruction as they take some time to kill the targeted locusts, but will limit their spread. The 2003-2005 locust upsurge ended due to lack of rain and colder temperatures which slowed down the breeding cycle, combined with intensive ground and aerial spraying operations which treated over 130,000km^2 at a cost of over US$400 million (FAO and WMO 2016).

Desert locust control is most effective before locust populations surge, and the FAO manages an international network of early monitoring, warning, and prevention systems in support of this goal (Zhang et al., 2019). While improvements in desert locust management have been largely effective in reducing the frequency of outbreaks (as seen in Figure B2), many challenges remain. Desert locust breeding areas are widespread and often in remote or insecure areas. Small breeding groups are easy to miss by monitors, and swarms can migrate quickly. In addition, control operations are slow and costly, resources for monitoring and control are limited outside of upsurges, and the cross-country nature of the thread creates coordination issues. Insecurity may also limit locust control activities (Showler and Lecoq, 2021).

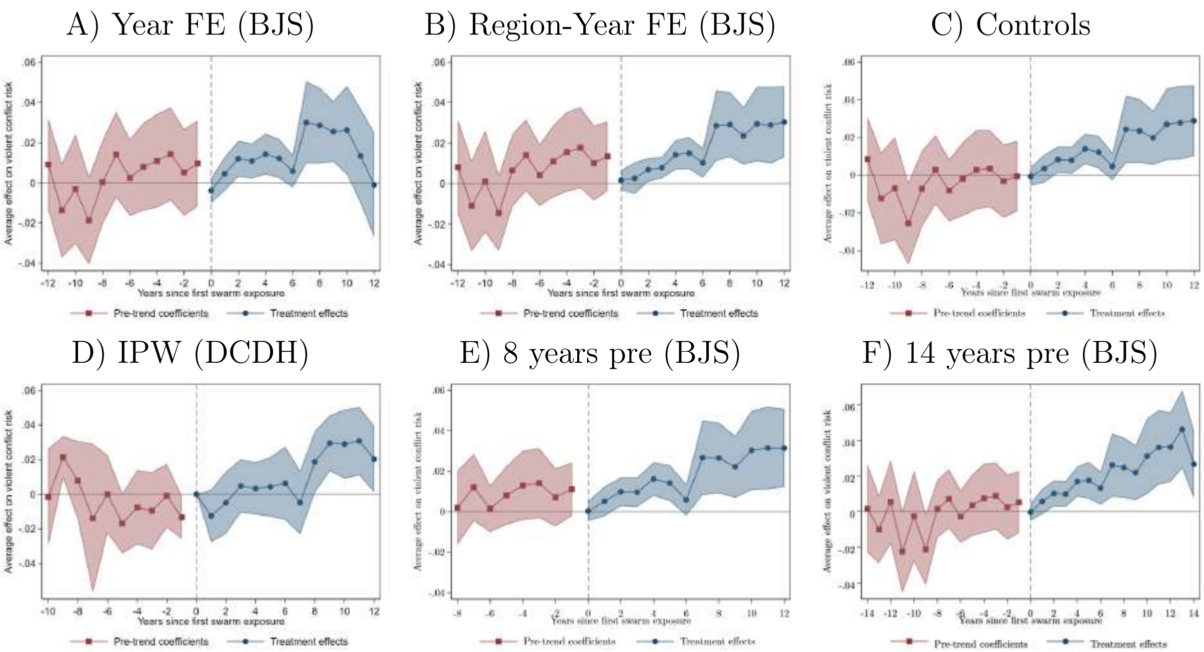
Appendix C: Robustness

Figure C1: Alternative locust swarm exposure event study estimators



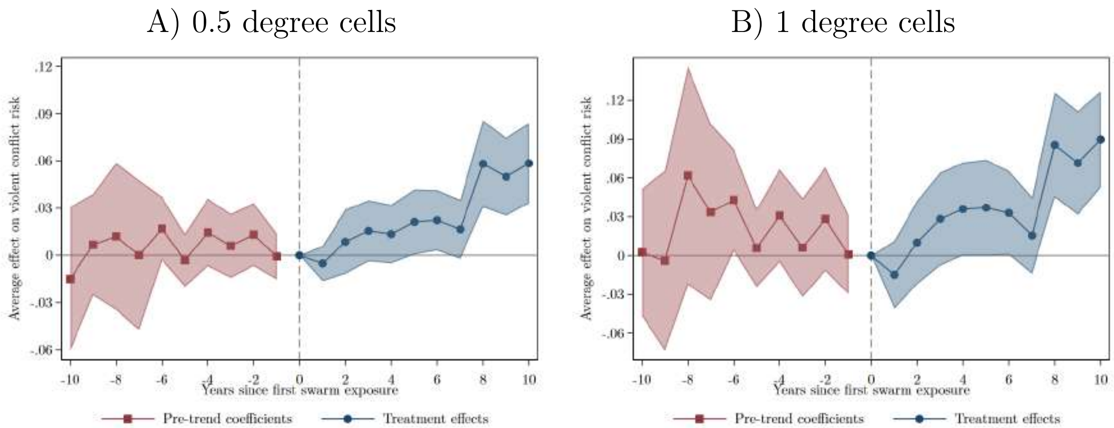
Each panel estimates dynamic effects of swarm exposure 'treatment' a different method: Borusyak et al. (2024), Callaway and Sant'Anna (2021), Cengiz et al. (2019), and De Chaisemartin and d'Haultfoeuille (2024). For each method I use a Stata package provided by the authors: *did_impute*, *cs_did*, *did_multiplegt_dyn*, and *stackedev*, respectively. The Callaway and Sant'Anna (2021) and De Chaisemartin and d'Haultfoeuille (2024) estimates *do not* include country-by-year fixed effects as the Stata packages do not allow for more flexible time fixed effects. The De Chaisemartin and d'Haultfoeuille (2024) Stata package only estimates a maximum of 10 pre-treatment effects, and treats the year before treatment as year 0. See the figure note for Figure 3 for additional detail on the estimation.

Figure C2: Sensitivity of locust swarm exposure event study results to different specifications



Each panel replicates Figure 3 but changes some aspect of the specification as indicated in the panel title. 'BJS' refers to the Borusyak et al. (2024) method. The main specification uses BJS with country-by-year fixed effects, no controls, and 12 years of pre-exposure treatment estimates. 'DCDH' refers to the De Chaisemartin and d'Haultfoeuille (2024) method, for which the Stata package natively allows the inclusion of weights and converged when including controls whereas the BJS package did not. 'IPW' refers to inverse propensity weights, constructed based on the estimated probability of being exposed to a locust swarm during the study period. See the figure note for Figure 3 for more detail.

Figure C3: Dynamic impacts of swarm exposure on violent conflict risk at different scales



Each panel replicates Figure C1 Panel C estimating dynamic impacts of locust swarm exposure of the risk of any violent conflict event using the De Chaisemartin and d'Haultfoeuille (2024) method at different spatial scales. The main analysis uses 0.25° cells. When collapsing to larger cells I take the maximum of the swarm exposure and violent conflict variables and the mean of control variables across smaller cells within the aggregate cell. See the figure note for Figure 3 for more detail.

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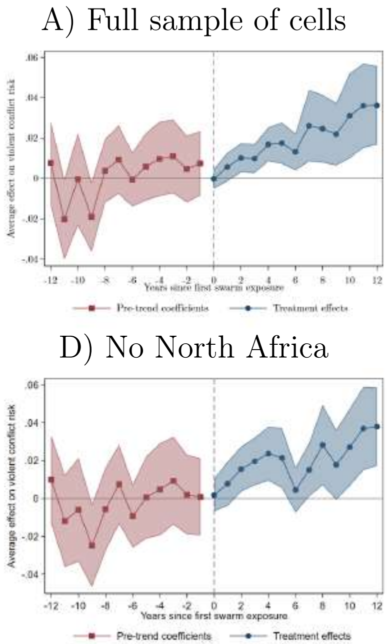
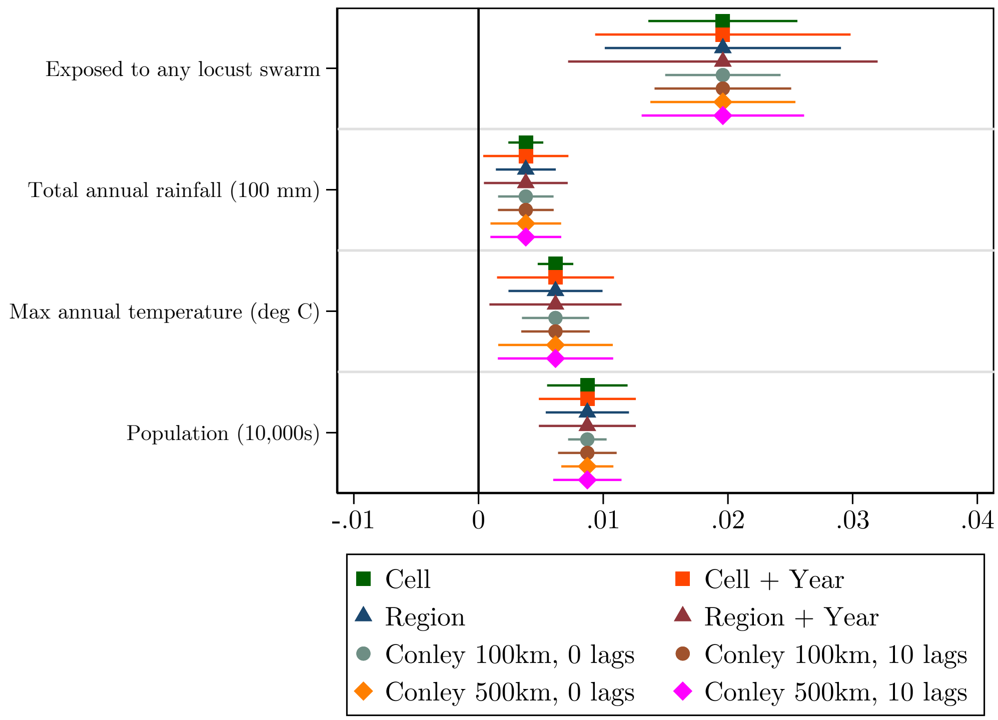
Figure C4: Sensitivity of locust swarm exposure event study results to different subsamples



Each panel replicates Figure 3 but changes the analysis sample as indicated in the panel title. The main analysis in Figure 3 excludes cells more than 100km from any locust swarm observation and outside the range of common support for the estimated probability of being exposed to a locust swarm in the study period across exposed and unexposed areas. 'SL2021' countries in panel B refers to the countries listed in Showler and Lecoq (2021) as areas where insecurity has limited desert locust control operations during the sample period. Panels C-F show results dropping specific geographic regions. See the figure note for Figure 3 for more detail.

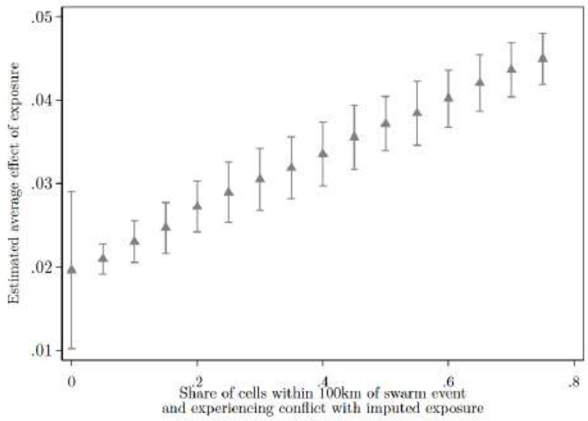
Figure C5: Estimated coefficients from Equation 1 with different SEs



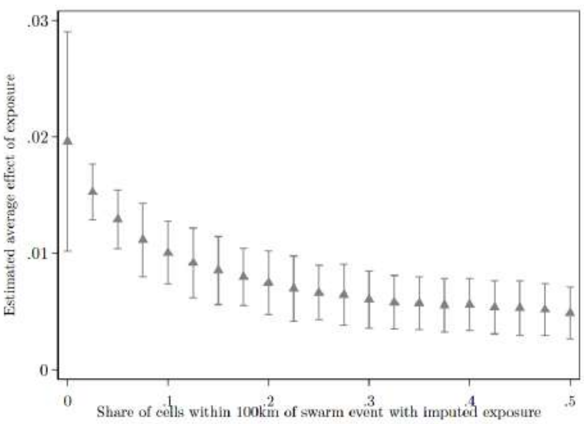
Note: The outcome variable is a dummy for any violent conflict observed. The figure shows 95% confidence intervals for estimates from Table A3 column (1) applying different clustering for the SEs. Observations are grid cells approximately 28×28km by year. Regressions also include country-by-year and cell FE.

Figure C6: Simulations of missing swarm observations

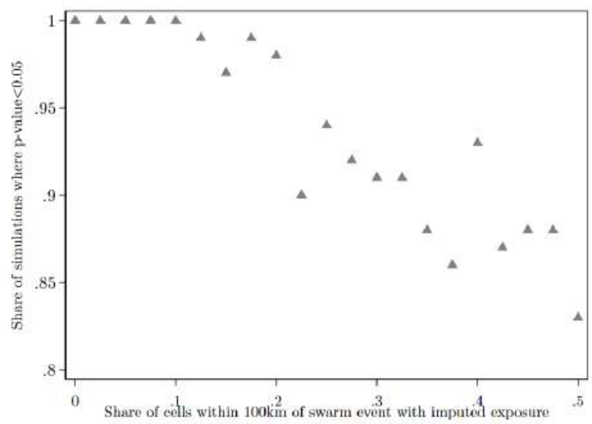
A) Betas, simulated swarms
in areas experiencing conflict
near swarm report



B) Betas, simulated swarms
in any area near swarm report

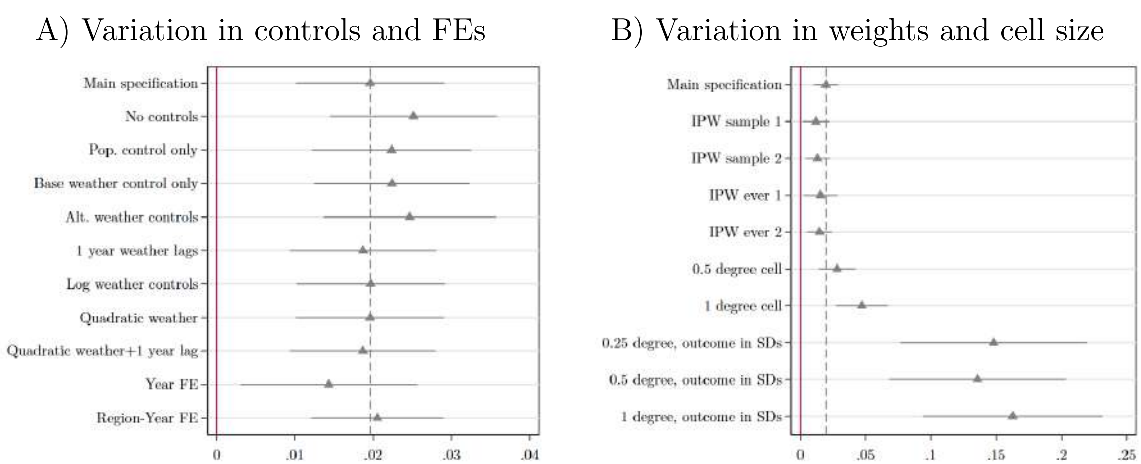


C) p-values, simulated swarms
in any area near swarm report



Note: The figures show the results from estimating Equation 1 in simulations imputing the presence of increasing shares of unreported locust swarms in cell-years with a locust swarm reported within 100km. For each share, I run 100 simulations randomizing which cell-years are imputed an unreported swarm, recalculating the swarm exposure treatment variable, and estimating the average impact of swarm exposure on violent conflict risk. In Panel A, I only impute swarms in cell-years both experiencing violent conflict and within 100km of a reported locust swarm, to simulate effects of missing swarm reports in insecure areas. In Panel B, I impute swarms across all cell-years within 100km of a reported locust swarm. Panels A and B report the average estimated effect across all simulations by share of imputed swarms, along with a 95% confidence interval for these estimates. Panel C reports the share of simulations where the p-value for the coefficient on swarm exposure is less than 0.05, by share of imputed swarms.

Figure C7: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative specifications



Note: Each estimate and 95% confidence interval is from a separate regression replicating Table A3 column 1 with a given modification. The dashed gray line indicates the main estimate from Table A3 column 1.

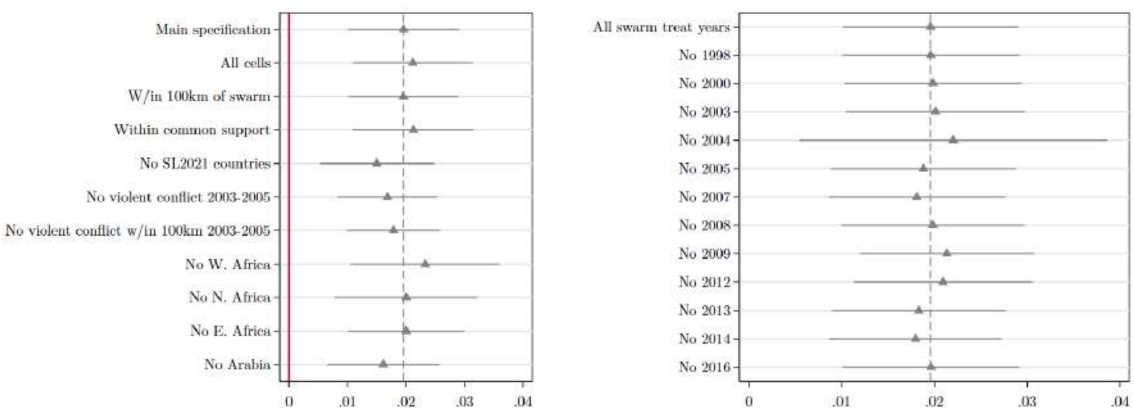
Panel A shows results varying controls and fixed effects. 'Alt.' weather controls replace the rainfall and temperature measures from WorldClim with measures from CHIRPS and ERA5, respectively.

Panel B shows results from applying weights to the regressions and increasing cell size from the base 0.25 degree cells. 'IPW' indicates inverse propensity weights based on the estimated probability of having been exposed to a locust swarm. I calculate propensity scores using a logit regression with a dummy for being exposed to a locust swarm on baseline cell characteristics and mean weather and country fixed effects. I calculate inverse propensity weights as $\frac{1}{p}$ for cells that were exposed and $\frac{1}{1-p}$ for cells that were not, where p is the estimated probability of swarm exposure. I assign cells with estimated probabilities outside the range of common support a weight of 0. 'Sample' weights are based on the probability of being exposed during the sample period, and 'ever' weights are based on the probability of being exposed at any point from 1985-2021. For both sets of weights, IPW 1 are the raw estimated weights and IPW 2 replaces weights above the 99th percentile with the 99th percentile weight.

For estimates in larger cells, I collapse the data by taking the maximum for swarm exposure and violent conflict dummies and means for rainfall, temperature, and population controls. I show effects on both a dummy for any violent conflict and on the standard deviation in the probability of any violent conflict, as the later approach maintains comparability in effect sizes as the baseline risk of any conflict in a cell-year increase with cell size.

Figure C8: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative samples

A) Variation in included cells B) Variation in included years



Note: Each estimate and 95% confidence interval is from a separate regression replicating Table A3 column 1 with a given modification. The dashed gray line indicates the main estimate from Table A3 column 1.
Panel A shows results varying the set of included cells. I first vary whether cells more than 100km from any swarm report and outside the range of common support of estimated swarm exposure probability are excluded. I then drop countries where Showler and Lecoq (2021) report insecurity prevented some locust control operations during the sample period and cells that experienced violent conflict during the 2003-2005 locust upsurge which might have prevented locust reporting. Finally I drop countries in particular geographic regions.
Panel B shows results from dropping individual years when locust swarm exposure events occurred.

Table C1: Average impacts of direct and spillover exposure to locust swarms on violent conflict risk

Outcome: Any violent conflict event	(1) All swarms	(2) 2003-2005 upsurge swarms
Exposed to swarm	0.020***	
	(0.005)	
Exposed to swarm w/in 100km outside cell	-0.002	
	(0.003)	
Exposed to swarm		0.018***
		(0.005)
Exposed to swarm w/in 100km outside cell		0.007*
		(0.004)
Observations	327646	327646
Outcome mean post-2004, no exposure	0.028	0.028
Country-Year FE	Yes	Yes
Cell FE	Yes	Yes
Controls	Yes	Yes

Note: The table presents results from estimating Equation 1 but including a spillover swarm exposure variable based on the first year a cell is within 100km of a swarm outside the cell. Column 1 considers all swarm exposure events while column 2 focuses on the 2003-2005 upsurge.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$